Engineering Project Management

ET 8401

Instructor: Kadigi (*Ph.D.*)

Email: ibrahim.kadigi@sua.ac.tz

0752-887-661

Why this course to you?

- This course introduces students to the principles, tools, and techniques of engineering project management, with a strong emphasis on telecommunication projects such as network deployment, fiber optic systems, mobile communication infrastructure, ICT systems, and software-based engineering projects.

- The course equips students with managerial, technical, and analytical skills required to plan, execute, monitor, and close engineering projects successfully while balancing scope, time, cost, quality, risk, and stakeholder expectations.

COURSE OUTLINE

Course Contents

- Introduction to Engineering Project Management
- Project Life Cycle and Organizational Structures
- Project Initiation and Feasibility Analysis
- Project Scope Management
- Project Time and Schedule Management
- Project Cost Management and Control
- Risk Management in Engineering Projects
- Quality Management in Engineering Projects
- Communication and Stakeholders Management
- Procurement and Contract Management
- Managing Change and Project Control
- Project Closure and Evaluation
- Project Management Software and Tools
- Professional Ethics, Leadership, and Sustainability

Course Contents

Introduction to Engineering Project Management

- Definition of a project and project management
- Characteristics of engineering projects
- Differences between engineering projects and routine operations
- Role of project management in telecommunication engineering
- Project constraints (scope, time, cost, quality)
- Overview of global project management standards (PMBOK, PRINCE2, Agile)

Course Contents

Project Life Cycle and Organizational Structures

- Project life cycle phases:
  - Initiation
  - Planning
  - Execution
  - Monitoring and controlling
  - Closure
- Phase gates and decision points
- Organizational structures:
  - Functional
  - Matrix
  - Projectized
- Roles and responsibilities of the project manager in telecom projects

Course Contents …

Project Initiation and Feasibility Analysis

- Project identification and selection
- Business case development
- Technical, economic, and financial feasibility
- Stakeholder identification and analysis
- Project charter:
  - Objectives
  - Scope
  - Assumptions and constraints
- Examples from telecom projects (e.g., 4G/5G rollout, fiber networks)

Course Contents …

Project Scope Management

- Defining project scope
- Work Breakdown Structure (WBS)
- Scope verification and validation
- Managing scope creep
- Change control procedures
- Case examples in telecom infrastructure projects

Course Contents …

Project Time and Schedule Management

- Activity definition and sequencing
- Network-based techniques:
  - Critical Path Method (CPM)
  - Program Evaluation and Review Technique (PERT)
- Gantt charts and milestones
- Float and slack calculations
- Schedule optimization techniques
- Application to telecom network deployment projects.

Course Contents …

Project Cost Management

- Types of project costs (direct, indirect, fixed, variable)
- Cost estimation techniques:
  - Analogous
  - Parametric
  - Bottom-up
- Budgeting and cost baseline
- Cost control mechanisms
- Introduction to Earned Value Management (EVM):
  - Planned Value (PV)
  - Earned Value (EV)
  - Actual Cost (AC)
  - CPI and SPI

Course Contents …

Risk Management in Engineering Projects

- Definition and types of risks:
  - Technical
  - Financial
  - Operational
  - Environmental
  - Regulatory
- Risk identification techniques
- Risk assessment and prioritization
- Risk response strategies:
  - Avoidance
  - Mitigation
  - Transfer
  - Acceptance
- Risk registers and monitoring
- Telecom project risk case studies

Course Contents …

Quality Management in Engineering Projects

- Concepts of quality in engineering
- Quality planning
- Quality assurance (QA)
- Quality control (QC)
- Standards and certifications (ISO, ITU standards)
- Quality management in telecom systems installation and testing

Course Contents …

Communication and Stakeholder Management

- Importance of communication in projects
- Communication planning
- Information flow and reporting
- Stakeholder analysis and engagement strategies
- Conflict management
- Managing multidisciplinary telecom project teams

Course Contents …

Procurement and Contract Management

- Make-or-buy decisions
- Procurement planning
- Tendering and supplier selection
- Contract types:
  - Fixed-price
  - Cost-reimbursable
  - Time and material
- Managing vendors and contractors
- Procurement challenges in telecom projects

Course Contents …

Managing Change and Project Control

- Change management process
- Configuration management
- Monitoring and controlling project performance
- Key performance indicators (KPIs)
- Project reporting and dashboards
- Corrective and preventive actions

Course Contents …

Project Closure and Evaluation

- Phase closure vs project closure
- Final deliverables and handover
- Documentation and archiving
- Post-implementation review
- Lessons learned
- Project success and failure analysis

Course Contents ...

Project Management Software and Tools

- Introduction to project management tools:
  - Microsoft Project
  - Primavera
- Open-source tools (e.g., GanttProject)
- Practical scheduling and resource allocation
- Use of software in telecom project planning

Course Contents …

Professional Ethics, Leadership, and Sustainability

- Ethics in engineering project management
- Professional responsibility and accountability
- Leadership styles in engineering projects
- Sustainability and environmental considerations
- Social impact of telecommunication projects

Introduction to Engineering Project Management

Introduction …

1.1 Meaning of a Project

- A project is a **temporary endeavor** undertaken to create a unique product, service, or result.
- Key characteristics of a project include:
  - **Temporary** – it has a definite start and end date
  - **Unique output** – produces a distinct deliverable
  - **Progressive elaboration** – details become clearer as the project advances
  - **Resource constrained** – limited by time, cost, and scope.
- Examples in Telecommunication Engineering:
  - *Installation of a fiber optic backbone in a city*
  - *Deployment of a 4G/5G mobile base station*
  - *Design and commissioning of a telecom data center*
  - *Implementation of a campus-wide Wi-Fi network*
- Routine activities like daily network monitoring are not projects, because they are ongoing and repetitive.

Introduction …

1.2 Meaning of Project Management

- **Project Management** is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.

- It involves:
  - Planning what must be done
  - Organizing resources
  - Leading teams
  - Monitoring progress
  - Controlling changes
  - Closing the project properly

- Engineering Perspective
  - In engineering, project management ensures that technical solutions are delivered efficiently while balancing cost, quality, time, and safety.

Introduction …

1.3 Importance of Project Management in Engineering

- Engineering projects are complex and risky. Project management helps to:

  1. Complete projects on time
  2. Control cost overruns
  3. Ensure technical quality
  4. Manage risks and uncertainties
  5. Coordinate multidisciplinary teams
  6. Meet client and regulatory requirements

- **Telecommunication Example**
  - Without proper project management:
    - A fiber network rollout may face delays due to poor scheduling
    - Equipment may arrive late due to procurement failures
    - Costs may escalate due to scope creep.

Introduction …

1.4 Project Constraints (Triple Constraint)

- Projects are constrained by:

  1. **Scope** – what work is included
  2. **Time** – project schedule
  3. **Cost** – budget
  4. **Quality** – performance standards (often added as the fourth constraint)

  - Change in one constraint affects the others

- **Examples**
  - If a telecom operator demands faster **rollout of 5G sites**, then:
    – Cost may increase (overtime, more staff)
    – Quality may be compromised if rushed
    – Scope may need adjustment.

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Introduction …

1.5 Engineering Projects vs Operations

Aspect	Engineering Project	Operations
Nature	Temporary	Ongoing
Output	Unique	Repetitive
Duration	Finite	Continuous
Example	Building a cell tower	Network maintenance

Introduction …

1.6 Characteristics of Engineering Projects

- High technical complexity
- Long planning horizon
- High capital investment
- Strict regulatory compliance
- Safety and environmental concerns
- Multidisciplinary teams

Telecom Case

- Rolling out submarine fiber cables requires:
  - Marine engineering
  - Environmental clearance
  - International coordination
  - Risk management

Introduction …

1.7 Role of Project Manager in Engineering Projects

- The Project Manager (PM) is responsible for:
  - Defining project objectives
  - Planning activities and schedules
  - Managing costs and resources
  - Coordinating engineers, vendors, and regulators
  - Managing risks and changes
  - Reporting progress to stakeholders

Telecom Project Manager Example

- In a Base Transceiver Station (BTS) installation project, the PM coordinates:
  - Radio frequency (RF) engineers
  - Civil engineers
  - Power suppliers
  - Equipment vendors
  - Local authorities

Introduction …

1.8 Project Management Standards and Methodologies

- Project management standards and methodologies provide structured frameworks, guidelines, and best practices for planning, executing, and controlling projects.

- They help project managers deliver projects efficiently, consistently, and successfully, regardless of project size or complexity.

  - PMBOK (PMI) – widely used globally
  - PRINCE2 – structured, process-driven
  - Agile – flexible, iterative (used in software & ICT)
  - Hybrid approaches – common in telecom projects

Introduction …

1.8 Project Management Standards and Methodologies …

PMBOK

- PMBOK, developed by the Project Management Institute (PMI), is a globally recognized standard that provides a comprehensive set of best practices, processes, and knowledge areas for project management.

- It is not a rigid methodology but a guideline that can be adapted to different types of projects.

Key features of PMBOK

- Organized into process groups (Initiation, Planning, Execution, Monitoring & Controlling, Closure)
- Covers **knowledge areas** such as scope, time, cost, risk, quality, procurement, and communication
- Highly flexible and applicable across industries.

- Telecommunication Example
  - A telecom company uses PMBOK principles to manage a national fiber-optic backbone project, applying formal scope management, cost control, risk analysis, and stakeholder communication processes.

Introduction …

1.8 Project Management Standards and Methodologies …

PRINCE2

- PRINCE2 is a structured, process-driven project management methodology developed in the UK. It emphasizes clear roles, responsibilities, documentation, and control at every project stage.

- It is not a rigid methodology but a guideline that can be adapted to different types of projects.

Key features of PRINCE2

- Divides projects into controlled stages
- Strong focus on business justification
- Clearly defined roles (Project Board, Project Manager, Team Manager)
- Suitable for large, regulated projects.

- Telecommunication Example
  - PRINCE2 is used by a government agency managing a national ICT infrastructure project, where strict reporting, approvals, and documentation are required at each phase.

Introduction …

1.8 Project Management Standards and Methodologies …

Agile Methodology

- Agile is a flexible and iterative approach to project management, commonly used in software development and ICT projects.

- It focuses on continuous improvement, customer collaboration, and rapid delivery of small functional components.

Key features of Agile

- Work is divided into short cycles called iterations or sprints
- Emphasizes adaptability and customer feedback
- Minimal documentation compared to traditional methods.

- Telecommunication Example
  - A telecom operator uses Agile to develop a mobile customer-care application, releasing new features every few weeks based on user feedback.

Introduction …

1.8 Project Management Standards and Methodologies …

Hybrid Project Management Approaches

- Hybrid approaches combine traditional methodologies (PMBOK or PRINCE2) with Agile practices, allowing flexibility while maintaining control.

- This approach is increasingly popular in telecommunication projects, which often include both hardware and software components.

Key features of Hybrid

- Traditional planning for infrastructure work
- Agile methods for software and system configuration
- Balances control with flexibility.

- Telecommunication Example
  - In a 5G network rollout project, civil works and tower construction are managed using PMBOK or PRINCE2, while network optimization software is developed using Agile methods.

Introduction …

1.8 Project Management Standards and Methodologies …

Summary Table

Methodology	Key Focus	Best Use Case	Telecom Example
PMBOK	Best practices & knowledge areas	Large, structured projects	Fiber network rollout
PRINCE2	Process control & governance	Government/regulatory projects	National ICT projects
Agile	Flexibility & rapid delivery	Software & ICT systems	Mobile apps, OSS/BSS
Hybrid	Balance of control & flexibility	Complex telecom projects	4G/5G deployment

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

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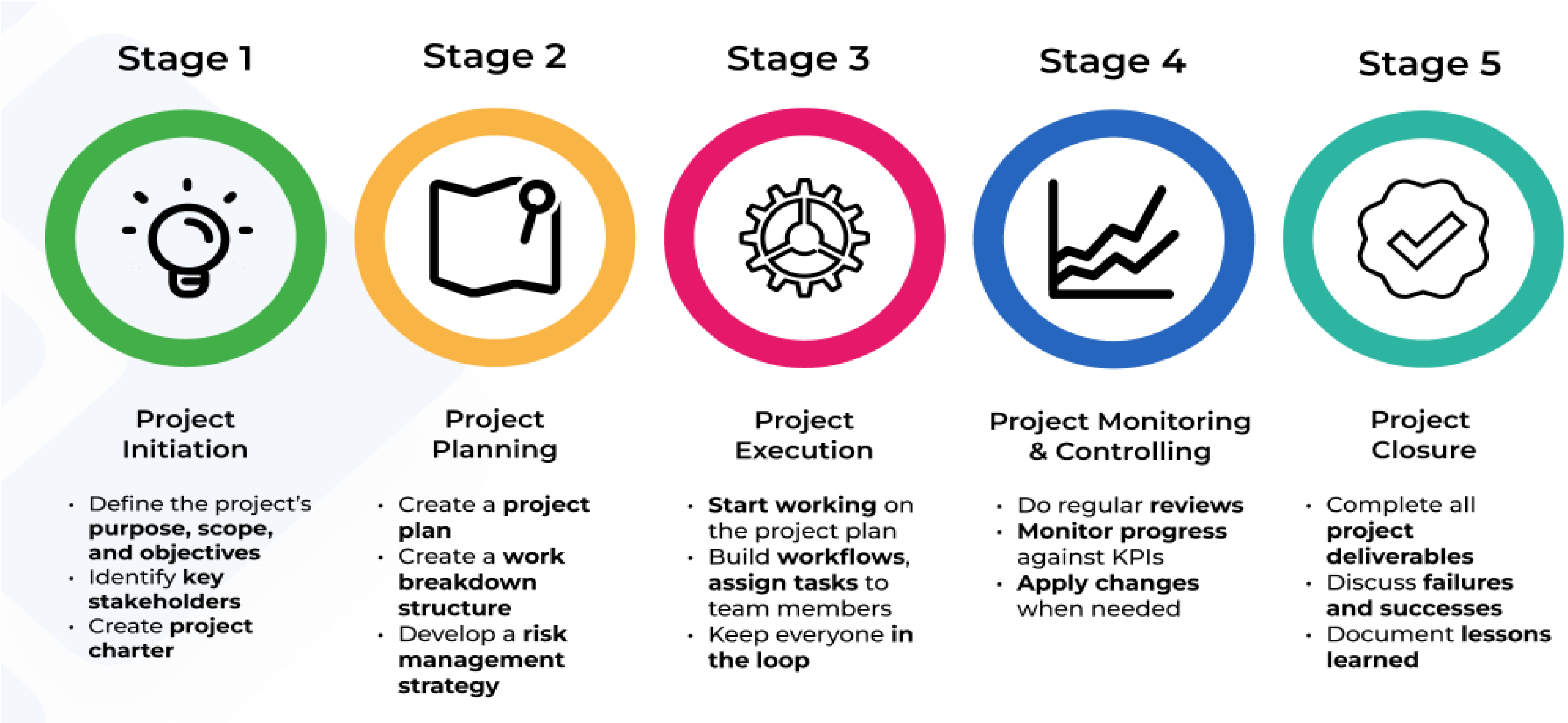
PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

2.0 Introduction

- Regardless of the industry in which a business operates or its primary priorities, project management processes include five stages: initiation, planning, execution, monitoring and controlling, and closure.
- Each stage has its own set of objectives, deliverables, and outcomes that are critical to the project's success.



PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

2.1 Meaning of a Project Life Cycle

- The project life cycle refers to the series of phases that a project passes through from start to finish.

- Main Phases:
  - **1.** Initiation
  - **2.** Planning
  - **3.** Execution
  - **4.** Monitoring and Controlling
  - **5.** Closure.

- Each phase has specific objectives and deliverables.

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Initiation Phase

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

1. Initiation Phase

- The project management process begins with the initiation stage.
- It's the phase when an idea becomes a meaningful business plan.
- This stage involves defining the project and its objectives.
- In other words, the main purpose of this phase is to define the project and obtain authorization to proceed.
- Before kicking off the project, there are a few questions that need to be answered:
  - What is the purpose of the project?
  - What are the project's objectives?
  - What kind of resources will be required?
  - Are there any potential constraints down the road?
  - How much time would it take for the project to be finished?
  - Does it have a budget range?
  - Who are the stakeholders?

PROJECT LIFE CYCLE ….

1. Initiation Phase …

- The purpose of this questionnaire is for managers to clearly outline the project's objectives and reasons through a set of formal documents.

- Some important PM documents include:
  - ***Business case:*** a business case explains how a business will benefit from this project. It also justifies various aspects, such as potential obstacles and expenses.
  - ***Feasibility study:*** a feasibility study assesses whether the project can be completed within a specified timeframe, considering various factors contributing to its successful completion.
  - ***Project charter:*** a project charter focuses on all necessary information about the project – what value it is going to deliver, what needs to be done, what is the expected timeline, what is the budget for the project, who is the assigned project manager, etc.

- Clarifying these questions and setting the ground basis for everyone involved is a crucial step for the project's success. Unclearly defined objectives and their rationale pose a threat, increasing the risk of project failure and failing to meet stakeholders' expectations.

PROJECT LIFE CYCLE ….

1. Initiation Phase …

Take this example:

- Suppose TTCL company initiates a project to:

  Expand fiber internet to rural districts.

- Outputs from this stage will be:
  - Business case
  - Preliminary cost estimates
  - High-level scope
  - Project manager appointment

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

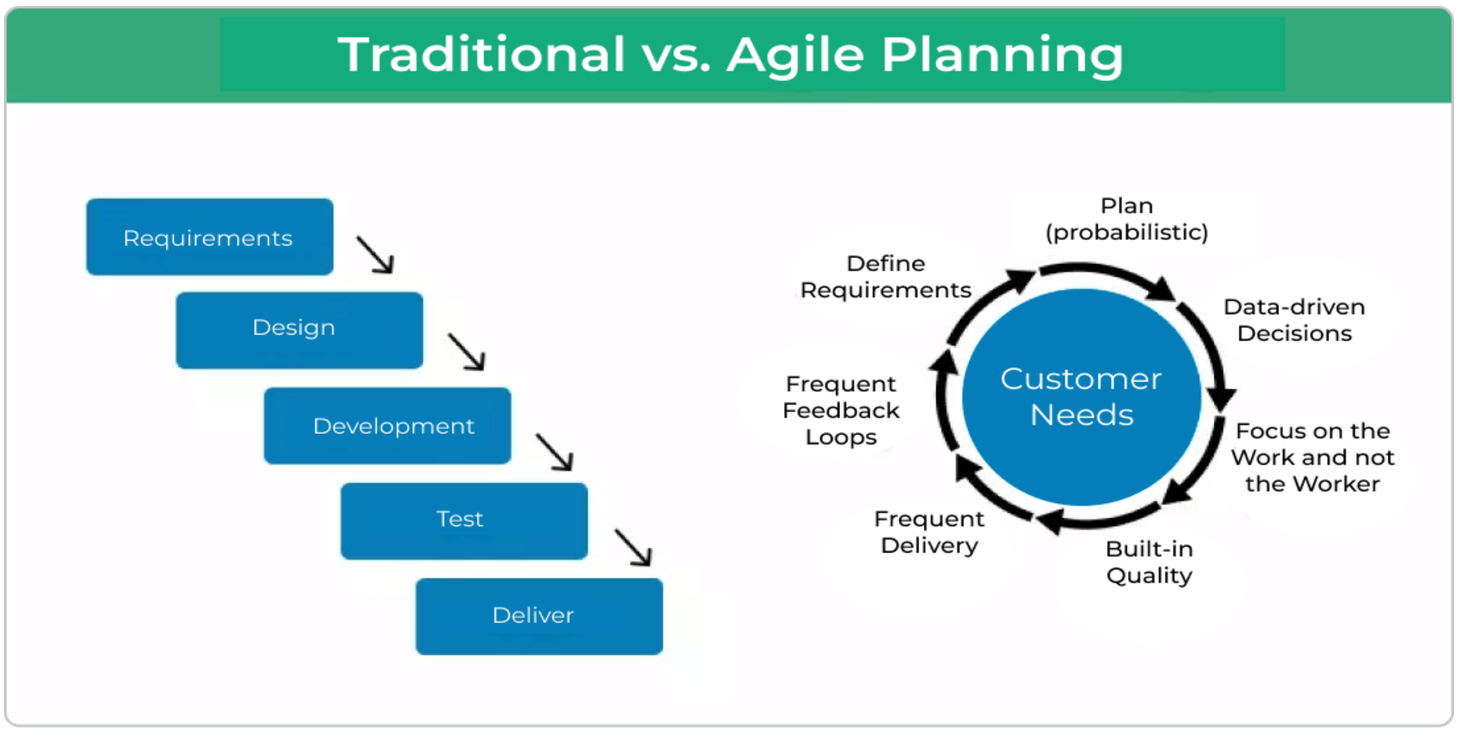
2. Project Planning

- Once the project is defined, it enters the planning stage.
- The primary goal of this stage involves creating a detailed project plan, which should include the following information:
  - setting project goals,
  - identifying the deliverables and creating a breakdown structure,
  - estimating the time and resources required to complete each task,
  - creating a project schedule
- During project planning, you should also create a risk management plan to outline the strategy if any risks occur associated with the project.
- In other words, this phase creates a detailed roadmap for the project execution

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

2. Project Planning …

- Whether you take a more traditional project planning approach or follow some of the Agile methodologies' guidelines and plan in an Agile way, setting achievable, specific, and measurable goals can be advantageous.
- To help with that, you can explore some goal-setting methods such as **S.M.A.R.T.** (Specific, Measurable, Achievable, Realistic, and Timely) or **C.L.E.A.R.** (Collaborative, Limited, Emotional, Acceptable, and Refined).



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PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

2. Project Planning …

- Key outputs
  - Project management plan
  - Scope definition and WBS
  - Schedule (CPM, Gantt charts)
  - Cost estimates and budget
  - Risk management plan
  - Communication plan

Telecom Example

- Planning a national LTE rollout involves:
  - Site acquisition plan
  - Equipment procurement schedule
  - Risk assessment (terrain, permits, power supply)

3. Project Execution

- After the planning stage comes execution, the backbone of a project's lifecycle.
- In this phase, you carry out the tasks and activities identified during planning.
- In other words, the main purpose of this stage/phase is to carry out the project plan and produce deliverables.
- During the project execution, project managers need to manage resources, address and resolve any shortcomings, build efficient workflows, and carefully monitor progress.

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

3. Project Execution …

- Moreover, maintaining open communication among project stakeholders ensures that everyone stays informed, promoting smooth project execution and staying on track:
- Balancing multiple responsibilities simultaneously can become overwhelming and hinder productivity.
- To keep up with everything going on, project leaders rely on project management tools to help them track work and collaborate seamlessly with everyone.
- Key activities, for example, if you are expanding fiber internet to rural districts will be:
  - Network construction (e.g., laying fiber cables)
  - Equipment installation (e.g., installing antennas)
  - Software configuration (e.g., configuring routers and switches,
  - Quality assurance (e.g., conducting site acceptance tests)
  - Team coordination

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

4. Project Monitoring and Controlling

- The distinction between the project monitoring phase and the project execution stage is subtle, as they often overlap.

- During monitoring, regular reviews of the project's status are essential to promptly identify and address issues, ensuring timely corrective actions are taken when needed.

- In other words, the main purpose of this phase is to track performance and ensure alignment with the plan.

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

4. Project Monitoring and Controlling ….

- Additionally, monitoring the project's progress against key performance indicators (KPIs) or OKRs (objectives and key results) informs if the project is going in the right direction and is on track.

- While in this stage, managers can still make all necessary adjustments to ensure the project is completed successfully and track the project timeline, budget, and quality requirements.

- In summary, key activities conducted here includes:
  - Schedule tracking
  - Cost control (EVM)
  - Quality inspections
  - Risk monitoring
  - Managing changes

4. Project Monitoring and Controlling ….

Take our previous example:

- If site construction is delayed:
  - PM revises schedule
  - Engages alternative contractors
  - Updates stakeholders

5. Project Closure

- At this stage, you want to ensure the stakeholder is satisfied with the project's outcomes.

- As a final to-do, you can conduct a last project review to identify any lessons learned and document the project results.

- The project closure phase marks the formal conclusion of a project and involves several key activities.

- Firstly, there is a comprehensive review of project deliverables against initial goals and objectives to ensure all requirements are met.

PROJECT LIFE CYCLE AND ORGANIZATIONAL STRUCTURES

5. Project Closure …

- Additionally, the closure phase involves releasing project resources, such as team members or equipment, and closing out financial accounts associated with the project.
- Handover of network to NOC (Network Operations Center)
  - Other activities involve documentation and lessons learned.
- The main purpose of this phase is to formally complete the project
- Final testing and certification is done here
- The project evaluation report is also produced here

Project Scope Management (PSM)

PSM

Project Scope Management

- PSM involves the processes required **to ensure that a project includes all the work required, and only the work required, to complete the project successfully**.

- It answers the fundamental question:
  - *What exactly will this project deliver, and what will it not deliver?*

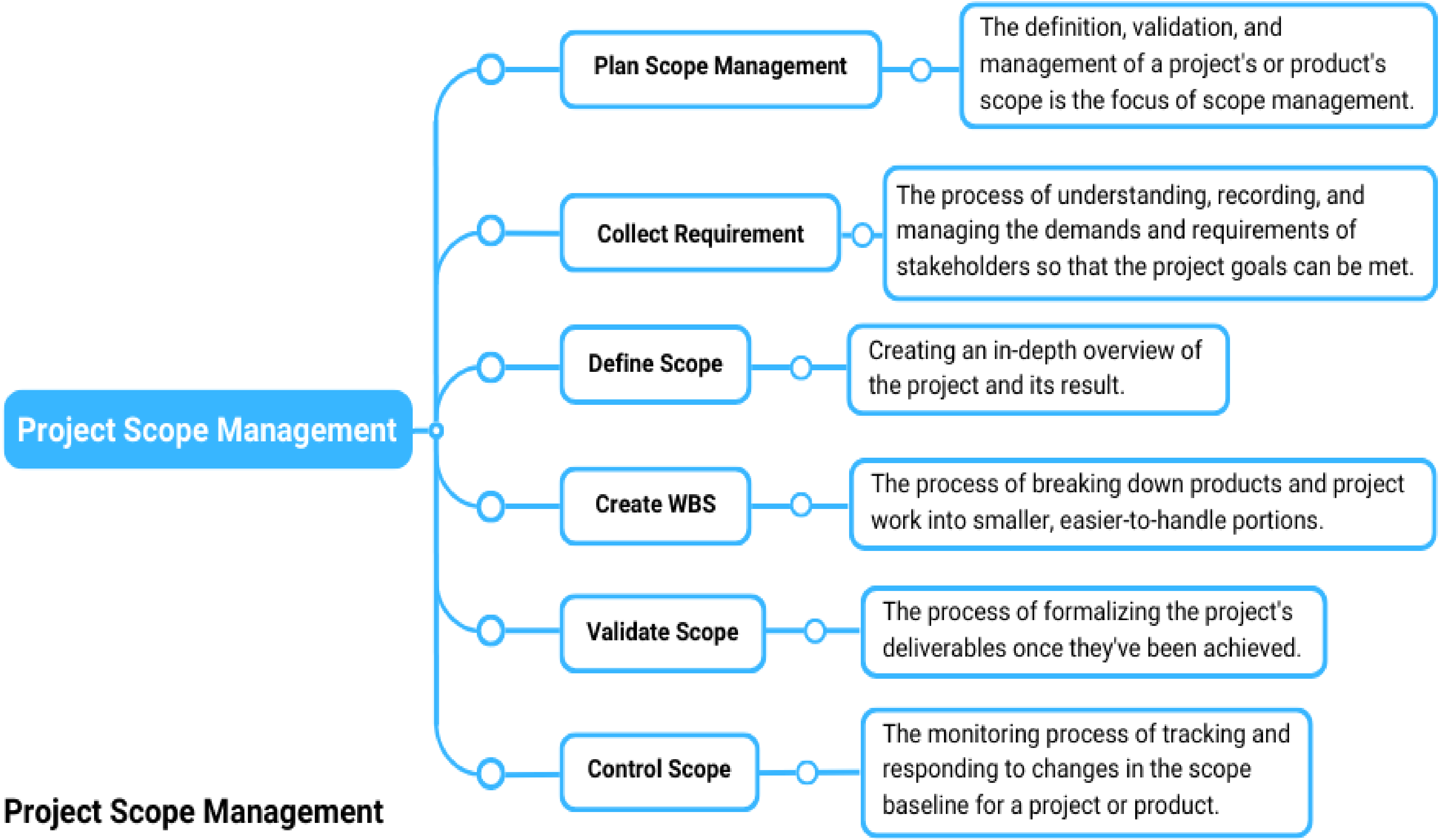
- Poor scope management is one of the leading causes of **project failure**, especially in large engineering and telecom projects.

PSM …

Importance of Scope Management

- **Effective scope management helps to:**
  - Prevent scope creep (uncontrolled expansion of project goals beyond their original)
  - Control cost overruns
  - Avoid schedule delays
  - Ensure stakeholder satisfaction
  - Improve accountability and clarity
- **In telecom projects, scope mismanagement can lead to:**
  - Installing more BTS sites than budgeted
  - Extending fiber length beyond planned routes
  - Adding new network features without funding.

PSM ...



Project Scope Management

PSM …

Defining Project Scope

- Project scope is a detailed description of:
  - Project objectives
  - Deliverables
  - Boundaries (inclusions and exclusions)
  - Constraints and assumptions
  - Acceptance criteria

- It is usually documented in a Scope Statement.

- The key elements of a Scope Statement include:
  - > Project Objectives
  - > Deliverables
  - > Inclusions
  - > Exclusions
  - > Constraints
  - > Assumptions
  - > Acceptance Criteria

PSM …

Defining Project Scope …

- The key elements of a Scope Statement include:
  - > Project Objectives (e.g., *To install a 50 km fiber optic backbone connecting five district offices.*)
  - > Deliverables
  - > Inclusions (e.g., *Trenching and duct installation; Fiber cable laying and splicing; Testing and commissioning*)
  - > Exclusions (e.g., *Customer last-mile connections; Maintenance after handover; Power backup systems*)
  - > Constraints (e.g., *Project must be completed within 6 months and a fixed budget*.)
  - > Assumptions
  - > Acceptance Criteria

PSM …

Work Breakdown Structure (WBS)

- A Work Breakdown Structure (WBS) is a hierarchical decomposition of the total project scope into smaller, manageable components called work packages.

If it is not in the WBS, it is not part of the project.

- The main Purpose of WBS includes:
  - Improves planning accuracy
  - Facilitates cost and time estimation
  - Clarifies responsibilities
  - Enhances monitoring and control

PSM …

Structure of a WBS

- **Level 1:** Project
- **Level 2:** Major deliverables
- **Level 3:** Sub-deliverables
- **Level 4:** Work packages

- Take our example of BTS Installation Project, how will the WBS look like:
  - **Level 1:** BTS Installation Project
  - **Level 2:**
    1. Site Acquisition
    2. Civil Works
    3. Tower Installation
    4. RF Equipment Installation
    5. Testing and Commissioning

PSM …

Structure of a WBS…

- **Level 3:**
  - ➢ Site clearing
  - ➢ Foundation excavation
  - ➢ Concrete casting
  - ➢ Curing

- **Level 4:**
  - ➢ Pouring foundation concrete (7 days)

PSM ...

Scope Verification and Validation

- Scope verification (validation) is the process of formally accepting completed project deliverables by the client or stakeholders.

- It answers questions like:

  Has the work been completed as agreed?.

- The main key activities
  - Inspections
  - Reviews
  - Testing
  - Client sign-off

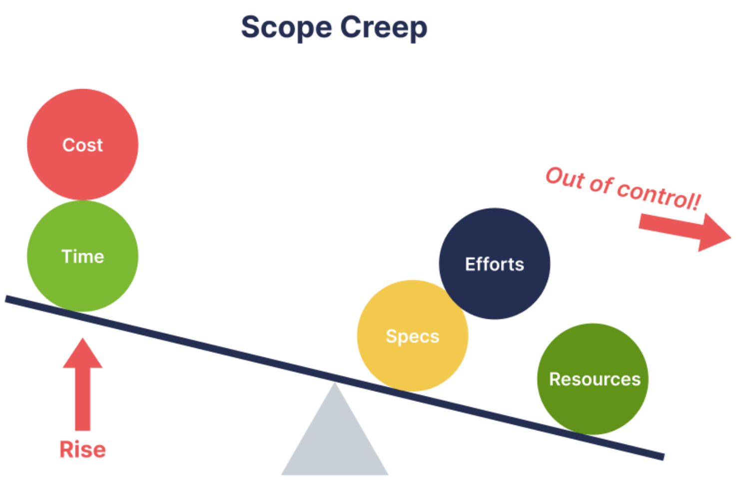
- For example, before accepting a fiber project:
  - Optical Time Domain Reflectometer (OTDR) tests are conducted
  - Signal loss is measured
  - Client signs off after meeting specifications

PSM …

Managing Scope Creep

- **Scope creep** refers to the uncontrolled expansion of project scope without corresponding adjustments in time, cost, or resources.

- Causes of Scope Creep
  - Poor initial scope definition
  - Informal client requests
  - Weak change control
  - Pressure from stakeholders
  - Technological evolution (common in telecom)



- Example of Scope Creep
  - Client requests additional BTS sites mid-project
  - Requests upgrade from 4G to 5G equipment
  - Adds redundancy links not in the original design

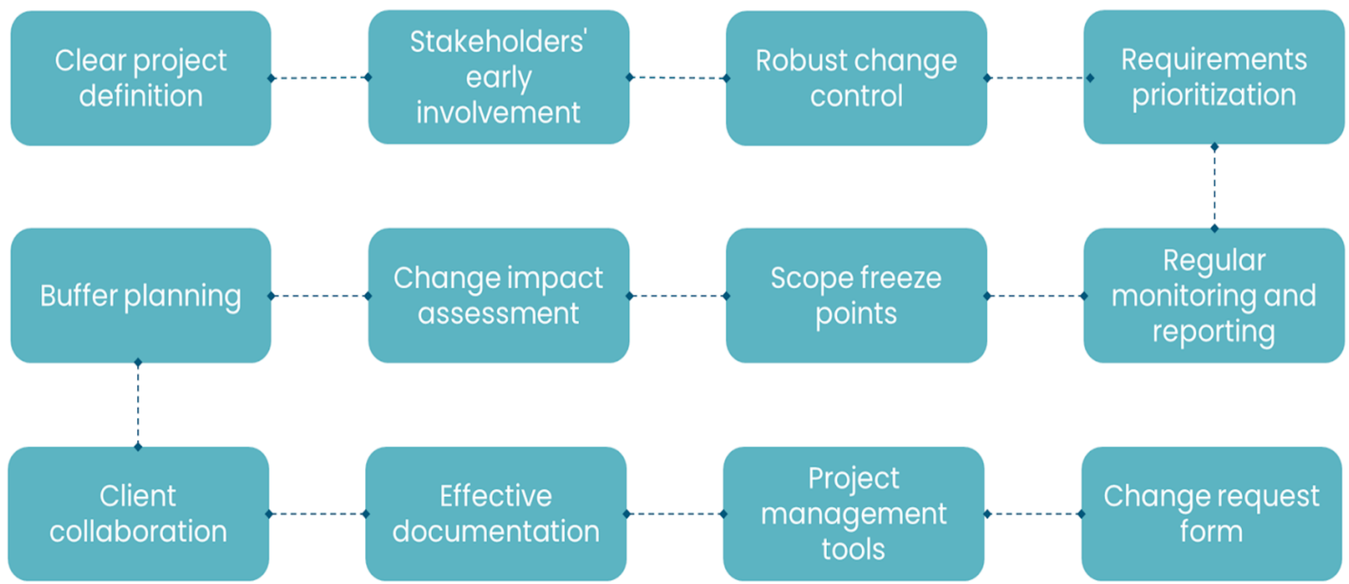
- Without proper control, these lead to cost overruns and delays.

PSM …

Managing Scope Creep…

Strategies to Control Scope Creep

- Clear scope statement
- Detailed WBS
- Formal change control process
- Strong stakeholder communication
- Regular scope reviews.

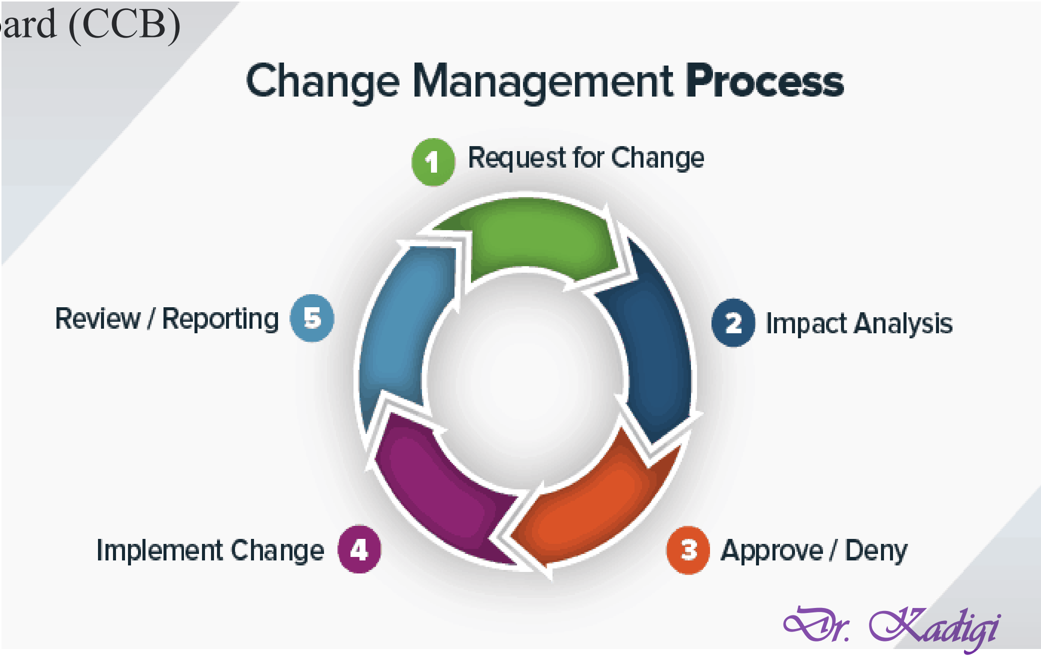


PSM …

Change Control Procedures

- **Change control** is a formal process used to review, approve, or reject changes to the project scope, schedule, or cost.

- Steps in Change Control Process
  1. Change request submission
  2. Impact analysis (time, cost, risk)
  3. Review by Change Control Board (CCB)
  4. Approval or rejection
  5. Update project documents
  6. Communicate decision.



PSM …

Change Control Procedures …

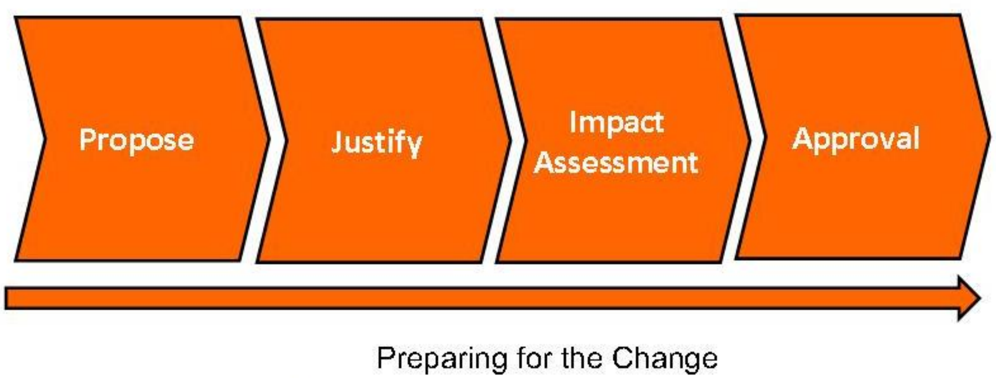
- **Take this example:**
- Suppose a telecom operator requests the addition of backup power systems at sites (while the project is in progress).

Impact Analysis:

- Cost increase: +15%
- Time extension: +1 month
- Additional procurement required

Change is approved only after:

- Budget revision
- Schedule update
- Contract amendment



Introduction …

Primary objective of PM

- The primary objective of project management is to ensure that a project is completed successfully within the defined scope, on time, and within budget while meeting or exceeding the stakeholders' expectations.

- This involves balancing the project constraints, which are often represented by scope, time, cost, quality, resources, and risk.

Introduction …

Primary objective of PM

- The primary objective of project management is to ensure that a project is completed successfully within the defined scope, on time, and within budget while meeting or exceeding the stakeholders' expectations.

- This involves balancing the project constraints, which are often represented by scope, time, cost, quality, resources, and risk.

Introduction …

Primary objective of PM ….

- **Scope:** Clearly defining and managing what is included and what is excluded from the project to meet the specific objectives laid out at the start.

- **Time:** Delivering the project's deliverables on schedule by effectively managing the time allocated for each activity and ensuring timely completion.

- **Cost:** Completing the project within the approved budget, managing finances efficiently throughout the project lifecycle to maximize resource allocation.

Introduction …

Primary objective of PM ….

- **Quality:** Ensuring that the project's outputs meet the necessary quality standards and stakeholder expectations, aligning with the project's objectives.

- **Resources:** Efficiently and effectively deploying project resources, including people, technology, and materials, to achieve the project goals.

- **Risk:** Identifying potential risks early on and managing them proactively to minimize their impact on the project.

Introduction …

- ICT Project Management is a specialized area of project management that requires precise oversight over complex interactions between software, hardware, network infrastructure, and data.

- It differs from general project management due to its highly technical elements and rapid pace of change in technology.

- ICT project management typically follows the traditional phases of project management but with a focus on technology-specific considerations.

- These phases include initiation, planning, execution, monitoring and controlling, and closure

Introduction…

Some examples of ICT projects are listed as follows:

- **Information Systems /Portals**
  - e-Government portals
  - City websites
  - Core banking information system
  - Immigration passport system
- **Network and Infrastructure**
  - Implementation of network (infrastructure) and equipment (routers, switches, etc)
  - Microsoft exchange email system
- **ICT Consultancy and Services**
  - e-Government ICT HR master plan
  - Network design and consultancy study
  - ICT master plan and e-Government roadmap.

Introduction…

Success and Failure ICT Related Projects

- ICT projects, like any complex undertakings, can either succeed spectacularly or fail disastrously.
- The reasons for success and failure in ICT projects can be numerous and varied, often intertwined with both technical and managerial factors.
- Understanding these reasons can help you or your organization better prepare and execute your ICT initiatives.

Introduction…

Reasons for ICT Project Success

❖ **Clear Goals and Objectives**

- Successful ICT projects have well-defined, realistic goals that align with business objectives, providing a clear roadmap and helping to keep the project on track.

❖ **Strong Project Management**

- Effective leadership and management, including meticulous planning, risk management, and resource allocation, are crucial.

- This also includes applying appropriate project management methodologies (like Agile or Waterfall, depending on the project needs).

Introduction…

Reasons for ICT Project Success …

❖ Stakeholder Engagement

- Regular communication with and active involvement of all stakeholders, including end-users, ensures that the project meets the actual needs of the organization and receives the necessary support.

❖ Competent Project Team

- Having a skilled and experienced team that can address the technical, management, and operational aspects of the project is vital.

- Continuous training and development are also key.

Introduction…

Reasons for ICT Project Success …

❖ **Adherence to Budget and Schedule**

- Successful projects stay within budget and meet their deadlines, which is often a result of effective scope and change management to prevent scope creep.

❖ **Quality Control Processes**

- Implementing robust quality assurance and testing processes ensures that the project meets the required standards and performs as expected.

❖ **Effective Change Management**

- Successfully managing the change that comes with new ICT implementations helps organizations smoothly transition to new systems and processes.

Introduction…

Reasons for ICT Project Failure

❖ Unclear or Shifting Project Objectives

- Lack of clear goals or changing objectives without proper adjustments can derail a project from meeting its intended outcomes.

❖ Poor Project Management

- Inadequate planning, underestimating the complexity of the project, poor risk management, and ineffective leadership can all lead to project failure.

❖ Lack of Stakeholder Engagement

- Insufficient involvement of end-users and other stakeholders can lead to a lack of buy-in and support, resulting in a product that does not meet the needs of its users.

Introduction…

Reasons for ICT Project Failure

❖ **Inadequate Resources**

- This can include insufficient budget, time constraints, or lack of necessary skills within the project team, all of which can impede project success.

❖ **Scope Creep**

- Uncontrolled changes or continuous growth in project scope without adjustments to time, cost, and resources can overwhelm the project team and lead to project failure.

Introduction…

Reasons for ICT Project Failure

❖ **Technical Issues**

- Underestimating technical challenges, choosing the wrong technologies, poor system integration, or inadequate testing can cause significant problems.

❖ **Resistance to Change**

- If the project does not address the change management adequately, users may resist adopting the new solution, thereby undermining the project's success.

Introduction…

How to Mitigate Risk and Enhance Project Success

- ❖ **Comprehensive Planning and Analysis**
  - Spend adequate time during the project initiation phase to plan thoroughly and perform detailed analysis.

- ❖ **Continual Monitoring and Evaluation**
  - Regularly check project progress against benchmarks and adapt strategies as needed.

- ❖ **Communication and Documentation**
  - Maintain open lines of communication across all levels of the project and document all decisions and changes thoroughly.

- ❖ **Invest in Training**
  - Equip your team and the end-users with the necessary knowledge and skills to handle the new systems effectively.

Project Management Foundation

Dr. Kadigi

Project Management Foundation

- Project management is a critical discipline used across various industries to plan, execute, control, and close projects.
- It involves the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.
- Understanding the foundation of project management provides a base for effective project execution and delivery.

Project Management Foundation

Foundational Elements of Project Management

➢ **1. Project Management Framework**

- Projects typically follow a sequential process, including initiation, planning, execution, monitoring and controlling, and closing (**project life cycle or phases**).

➢ **2. Core Functions**

- Ensuring the project includes all the work required, and only the work required, to complete the project successfully (***scope Management***).

- Developing and managing a timetable that ensures the project is completed on time (***Schedule and Time Management***).

- Planning and managing the budget to ensure the project is completed within the approved expenditure (***cost management***).

Project Management Foundation

Foundational Elements of Project Management

- **2. Core Functions….**
  - Determining quality policies, objectives, and responsibilities so that the project will satisfy the needs for which it was undertaken (*quality management*).
  - Identifying, acquiring, and managing resources needed for the successful completion of the project (*resource management*).
  - Acquiring or procuring goods and services from outside sources to meet project requirements (*Procurement Management*).
  - Engaging all stakeholders appropriately, and managing their expectations and satisfaction (*stakeholder management*).

3. Project Composition and Responsibilities

- The composition required to carry out the project management shall consist of the following parties:
  - Project steering committee
  - Project Manager
  - Project Team

Project Management Foundation

- 3. Project Composition and Responsibilities …
- **Project Steering Committee**
  - Management group chaired by a senior executive as the Project Sponsor

Main Responsibilities of the Project Steering Committee

- Provide management guidance and feedback
- Provide access to resources (personnel, information, etc.)
- Review all recommendations on opportunities, approaches and resources
- Maintain focus on business needs
- Ensure adequate funding is available to support the project
- Make decisions concerning resource conflicts.
- Ensure risk assessment is regularly reviewed and the risks are managed

Project Management Foundation

- **3. Project Composition and Responsibilities …**
- **Project Manager**
  - A person assigned by the Project Steering Committee for day-to-day management on their behalf

Main Responsibilities of members of the Project Manager

- Select the project team, together with the sponsor
- Maintaining a close working relationship with the sponsor
- Identify and manage the project stakeholders
- Identify and manage the risks
- Allocate and secure resource commitments
- Monitoring and tracking the project progress
- Solving problems that interfere with progress
- Controlling cost
- Leading the project team
- Informing stakeholders of progress
- Delivering the project deliverables on time
- Manage the performance of the project team

Project Management Foundation

➢ 3. Project Composition and Responsibilities …

- **Project Team**
  - Individuals selected to achieve the deliverables

Main Responsibilities of the Project Team

- Helping the Project Manager to deliver the objectives
- Participate in team meetings, visits, and workshops
- Use their competencies to carry out the tasks they have been given
- Alerting the Project Manager to any risks that appear
- Provide information for the project documentation as needed

Project Management Foundation

- **4. Project Management Tools**

- **Project Management Software**
  - Tools like Microsoft Project, Asana, and Trello help in scheduling, resource allocation, and budget management.

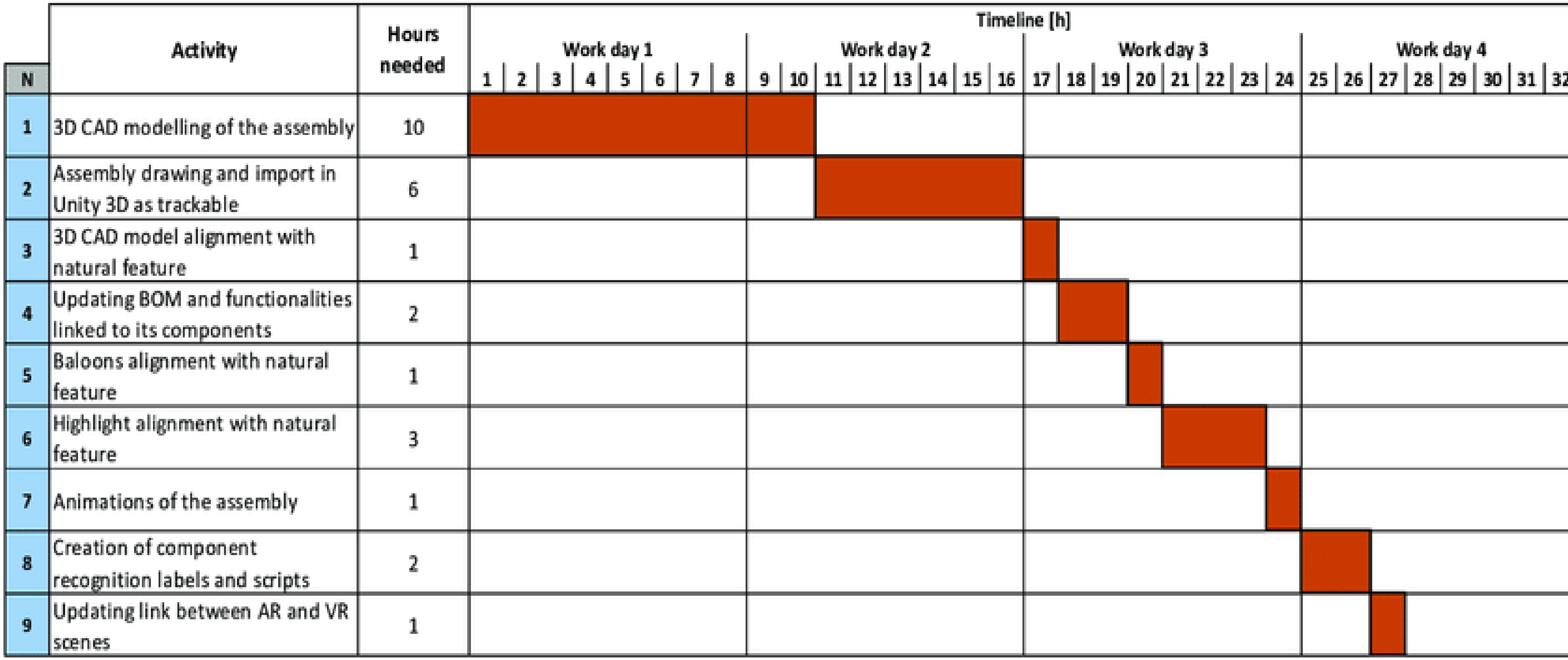
- **Gantt Charts**
  - Visual representations of the project schedule where activities, along with their start and end dates, are shown against a calendar.

- **Critical Path Method**
  - A step-by-step project management technique to identify activities on the critical path.

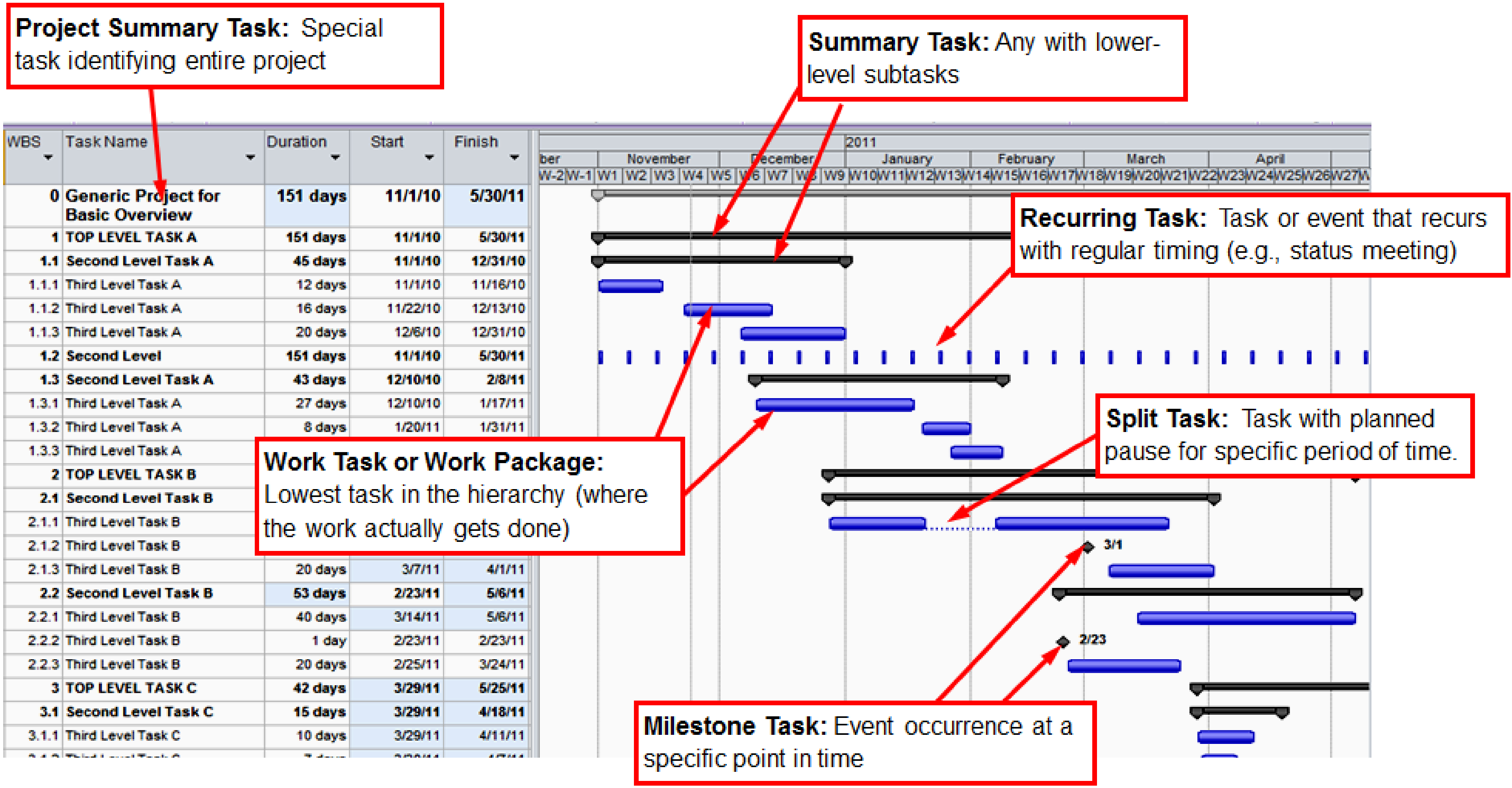
  - It is an approach to project scheduling that determines the longest sequence of dependent tasks and gauges the minimum amount of time needed to complete them.

- **Earned Value Management**
  - A technique that combines scope, schedule, and resource measurements to assess project performance and progress.

Gantt Charts

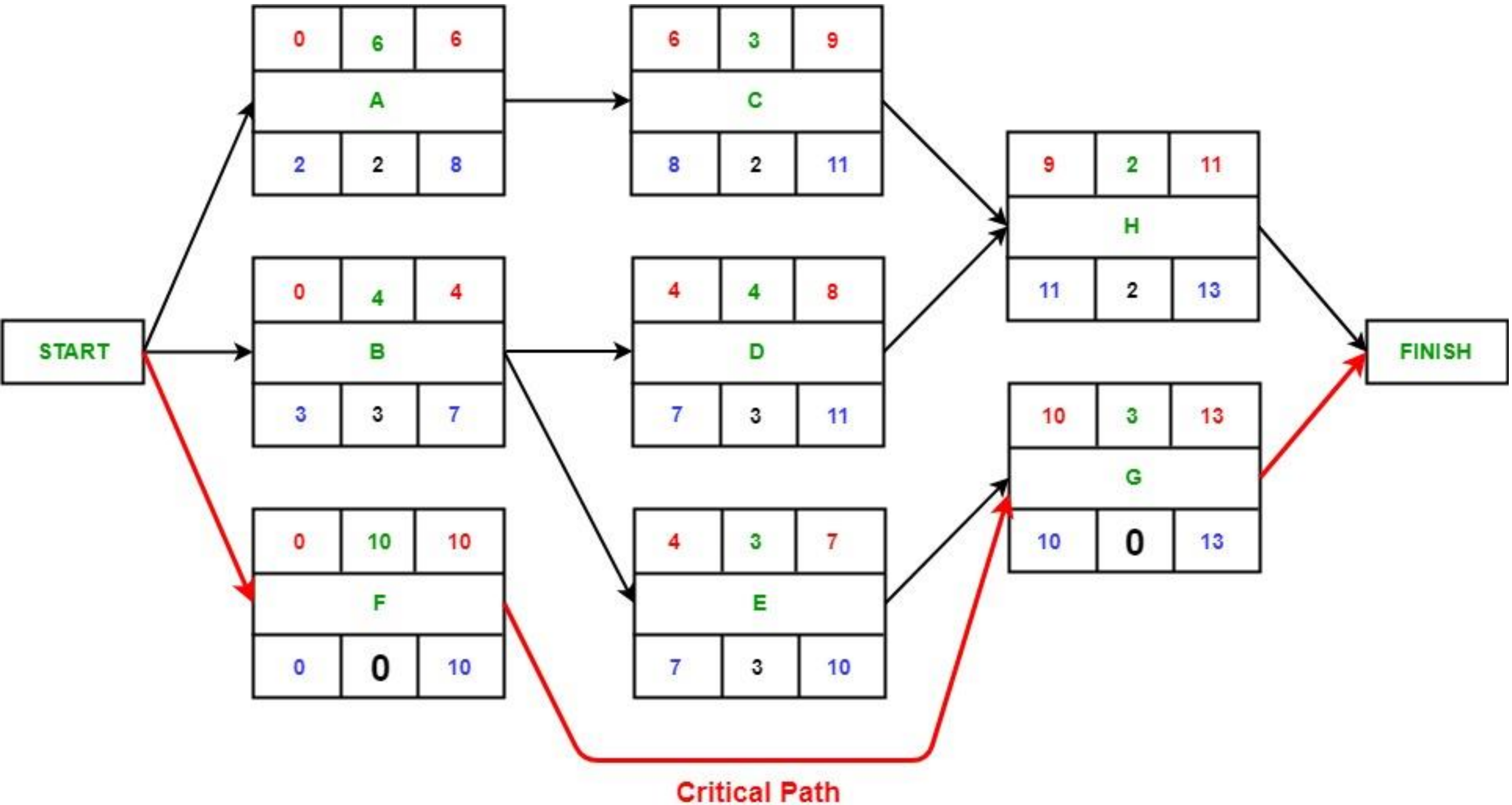


Gantt Charts



WBS	Task Name	Duration	Start	Finish
0	Generic Project for Basic Overview	151 days	11/1/10	5/30/11
1	TOP LEVEL TASK A	151 days	11/1/10	5/30/11
1.1	Second Level Task A	45 days	11/1/10	12/31/10
1.1.1	Third Level Task A	12 days	11/1/10	11/16/10
1.1.2	Third Level Task A	16 days	11/22/10	12/13/10
1.1.3	Third Level Task A	20 days	12/6/10	12/31/10
1.2	Second Level	151 days	11/1/10	5/30/11
1.3	Second Level Task A	43 days	12/10/10	2/8/11
1.3.1	Third Level Task A	27 days	12/10/10	1/17/11
1.3.2	Third Level Task A	8 days	1/20/11	1/31/11
1.3.3	Third Level Task A			
2	TOP LEVEL TASK B			
2.1	Second Level Task B			
2.1.1	Third Level Task B			
2.1.2	Third Level Task B			
2.1.3	Third Level Task B	20 days	3/7/11	4/1/11
2.2	Second Level Task B	53 days	2/23/11	5/6/11
2.2.1	Third Level Task B	40 days	3/14/11	5/6/11
2.2.2	Third Level Task B	1 day	2/23/11	2/23/11
2.2.3	Third Level Task B	20 days	2/25/11	3/24/11
3	TOP LEVEL TASK C	42 days	3/29/11	5/25/11
3.1	Second Level Task C	15 days	3/29/11	4/18/11
3.1.1	Third Level Task C	10 days	3/29/11	4/11/11

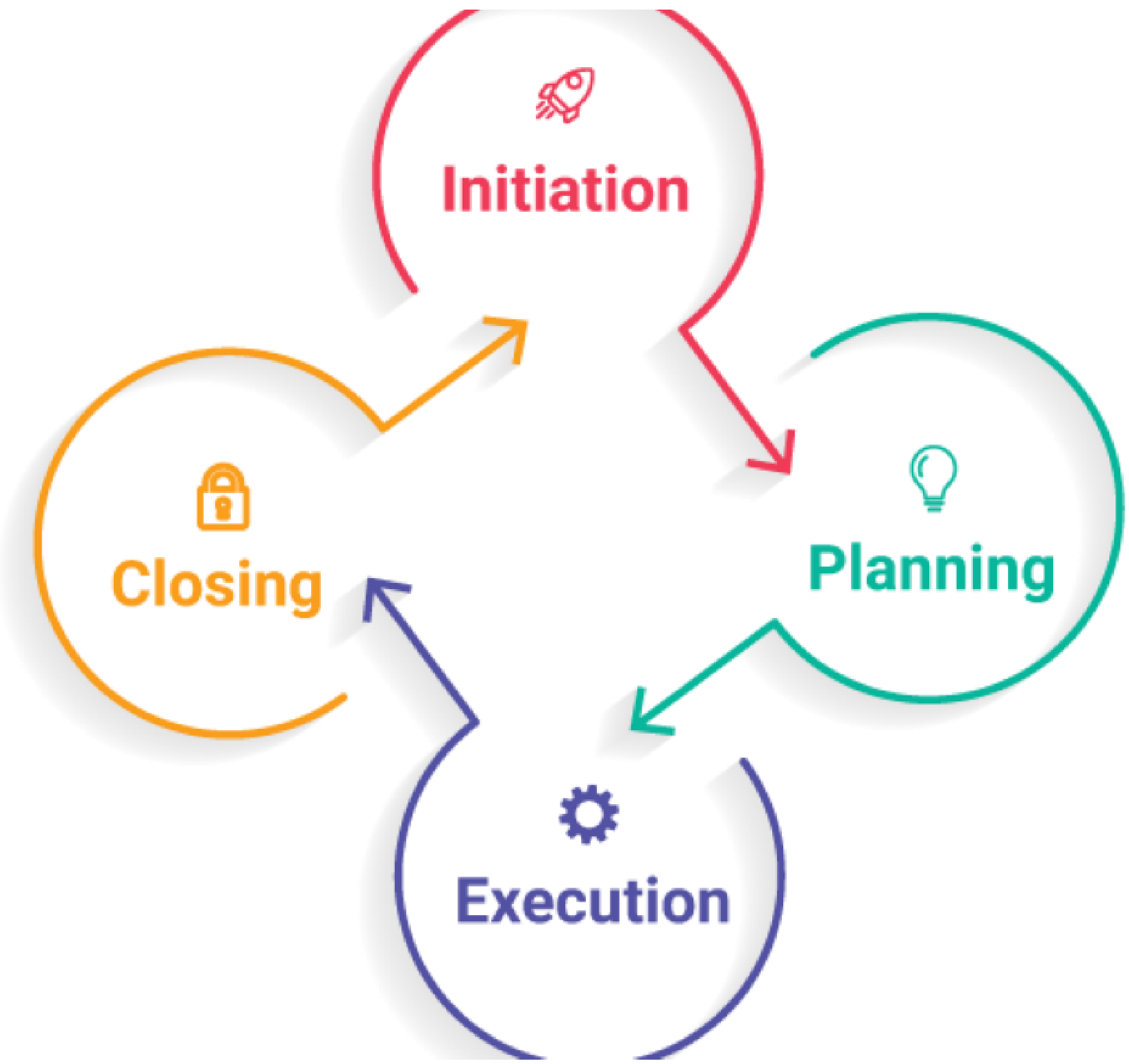
Critical Path



Project Life Cycle

Project Life Cycle

- The project life cycle (PLC) is a fundamental concept in project management that describes the stages through which a project progresses from **start** to **finish**.
- It provides a structured approach to planning and executing projects, ensuring consistency, clarity, and control throughout the entire process.
- The life cycle typically comprises four primary phases: **initiation**, **planning**, **execution**, and **closure**.
- Each phase has specific objectives and tasks that contribute to the overall success of the project.


Initiation
Planning
Execution
Closing

Project Life Cycle

Why is the Project Life Cycle important?

- Understanding and following the project life cycle is crucial for project managers and teams because it provides:
  - A structured approach that helps to manage large and complex projects by breaking them into manageable phases.
  - Efficiency through planned stages, resources and efforts are directed appropriately, reducing waste and increasing productivity.
  - Improved control because each phase has its own process and deliverables, making it easier to monitor progress, make adjustments and manage risks effectively.

Initiation Phase

PLC Initiation Phace

Initiation Phase

- The initiation phase is the starting point of a project.
- During this phase, the project's feasibility and relevance are evaluated to ensure that it should proceed.
- This phase involves defining the project at a broad level.
- It often starts with a business case that explains the need for the project and its expected benefits.
- A feasibility study may be conducted to assess the practicality and value of the project.
- Key project roles such as the project sponsor, project manager, and other critical stakeholders are identified.

PLC – Initiation Phase

Initiation Phase Key Activities

Identifying the Project's Purpose and Scope

- This involves defining what the project is supposed to accomplish and the boundaries within which it must operate.
- The Project Objective Statement (POS) describes in concrete terms what the project aims to achieve.
- It articulates the final outcome of the project and how the success of the project will be measured.
- **Project Scope** is the boundary that defines where the project begins and where it ends.

PLC – Initiation Phase

Initiation Phase Key Activities …

Developing the Project Charter

- This document formally authorizes the existence of the project and provides the project manager with the authority to apply organizational resources to project activities.

Stakeholder Identification

- Recognizing all parties affected by the project and understanding their influences and interests relative to the project's objectives.

PLC – Initiation Phase

Example of Initiation Phase

- Let's consider an example of the initiation phase in the context of launching a new **software development project for a mobile banking application for CRDB Bank**.

- This example will illustrate the typical activities and decisions made during the initiation phase of a project.

PLC – Initiation Phase

Example of Initiation Phase

Project: Mobile Banking Application Development

Background

A regional bank wants to develop a mobile banking application to expand its services and provide customers with a convenient, secure, and reliable mobile banking experience. The bank aims to increase customer satisfaction and compete with other financial institutions that offer advanced digital banking solutions.

PLC – Initiation Phase

Example of Initiation Phase …

Step 1: Project Ideation and Proposal

Business Case

A business case is developed by the CRDB bank's digital services department outlining the need for a mobile banking app to enhance customer engagement, reduce operational costs, and increase revenue through new digital products.

Feasibility Study

A preliminary study assesses the technical feasibility, market demand, and financial justification for the project.

This study includes an analysis of the bank’s current IT infrastructure, potential technology partners, and a cost-benefit analysis.

PLC – Initiation Phase

Example of Initiation Phase …

Step 2: Stakeholders Identification

Internal Stakeholders

Include the bank's executive team, IT department, customer service, marketing, and compliance teams.

External Stakeholders

Potentially involve software development firms, cybersecurity consultants, regulatory bodies, and third-party service providers (e.g., payment processors).

PLC – Initiation Phase

Example of Initiation Phase …

Step 3: Define Project Objective

Objectives

Develop a user-friendly mobile application that allows customers to perform transactions, check balances, apply for loans, and receive notifications about their accounts.

Scope

The project will include app design, development, integration with existing banking systems, compliance checks, user testing, and initial rollout in selected regions.

PLC – Initiation Phase

Example of Initiation Phase …

Step 4: Project Charter Development

Document Creation

A project charter is prepared, detailing the project's purpose, objectives, scope, key deliverables, milestones, high-level timeline, and budget. It also outlines the roles and responsibilities of the project team.

Authorization

The project charter is reviewed and approved by the bank's board of directors, granting formal authorization to proceed with the project.

PLC – Initiation Phase

Example of Initiation Phase …

Step 5: Appoint a Project Manager and Form a Project Team

Project Manager Appointment

A project manager with experience in digital projects and software development is appointed.

Team Formation

The project team is assembled, including internal IT staff, software developers, user experience (UX) designers, and representatives from marketing and compliance.

PLC – Initiation Phase

Example of Initiation Phase …

Step 6: Initial Stakeholders Meeting (Kickoff Meeting)

Meeting Goals

To align the vision, discuss the project charter, and ensure all stakeholders understand their roles and the project's expectations.

Communication plan

Establishing how communication will be managed throughout the project, including reporting formats, meeting schedules, and escalation procedures.

PLC – Initiation Phase

Example of Initiation Phase …

Concluding Remarks of Initiation Phase:

By the end of the initiation phase, the project has a clear direction, with a defined scope and objectives, and a committed team ready to move into the planning phase.

This sets the groundwork for detailed planning and execution, ensuring everyone involved has a unified understanding of the project's goals and expectations.

Planning Phase

PLC – Planning Phase

Planning phase

- The planning phase involves detailing the path from project start to completion.
- It focuses on developing a plan to guide the team through the execution and closure phases of the project.
- This includes developing project management plans for scope, schedule, cost, quality, resources, communications, risk, procurement, and stakeholder engagement.
- Setting objectives and key performance indicators (KPIs)
  - Specific, measurable, achievable, relevant, and time-bound objectives are set to ensure that project goals are clearly understood and measurable.
- Identifying potential risks and devising strategies to mitigate them is crucial for preventing project disruption

PLC – Planning Phase

An Example of a Planning Phase (continue with our previous example)

Step 1: Project Management Plan Development

Comprehensive Planning

The project manager leads the development of a comprehensive project management plan, which includes integrated plans for scope, schedule, costs, quality, resources, communication, risk, procurement, and stakeholder engagement.

Scope Management Plan

This document outlines how the project scope will be defined, validated, and controlled. It ensures that all required work (and only the required work) is included in the project.

Schedule Management Plan

Defines the project timeline, key milestones, dependencies, and the critical path. Tools like Gantt charts or project management software might be used to visualize the timeline.

Cost Management Plan

Includes budget estimates, funding allocations, and cost control measures to keep the project within the approved budget.

PLC – Planning Phase

An Example of a Planning Phase …..

Step 1: Project Management Plan Development

Quality Management Plan

Establishes quality benchmarks, testing protocols, and review processes to ensure the app meets all technical requirements and user expectations.

Resource Management Plan

Details the human, technological, and physical resources needed for the project, including how they will be acquired, allocated, and managed.

Communication Management Plan

Specifies the communication methods, frequencies, and protocols for updating stakeholders and managing expectations throughout the project.

PLC – Planning Phase

An Example of a Planning Phase ….

Step 1: Project Management Plan Development

Risk Management Plan

Identifies potential risks, assesses their impact and likelihood, and outlines mitigation strategies to manage risks proactively.

Procurement Management Plan

If external vendors or contractors are needed, this plan outlines how they will be selected and managed.

Communication Management Plan

Identifies all project stakeholders and outlines strategies for engaging them effectively throughout the project lifecycle.

PLC – Planning Phase

An Example of a Planning Phase …

Step 2: Define Project Milestones and Deliverables

Milestones

Establish key milestones such as completion of app design, end of coding phases, testing stages, and app launch.

Deliverables

Define specific outputs for each phase, such as design prototypes, beta versions of the app, final app builds, and user manuals.

PLC – Planning Phase

An Example of a Planning Phase ….

Step 3: Assign Roles and Responsibilities

Team Roles

Clearly define the roles and responsibilities of each team member to ensure accountability.

This includes delineating who is responsible for design, development, testing, and deployment.

Vendor and Stakeholder Roles

Outline the responsibilities of any third-party vendors or external stakeholders, including what deliverables they are responsible for.

PLC – Planning Phase

An Example of a Planning Phase ….

Step 4: Risk Assessment and Mitigation Strategies

Risk Identification

Conduct workshops and meetings to identify potential risks.

Common risks might include technical challenges, delays in delivery, security vulnerabilities, and regulatory compliance issues.

Mitigation Strategies

Develop strategies to address these risks, such as adopting agile development methodologies, conducting regular security audits, and ensuring compliance with financial regulations.

PLC – Planning Phase

An Example of a Planning Phase ….

Step 5: Establish Budget and Funding Requirements

Budgeting

Create a detailed budget that includes estimates for labor, software, hardware, third-party services, and contingency funds.

Funding Approval

Present the budget to stakeholders for approval and secure the necessary funding to move forward.

PLC – Planning Phase

An Example of a Planning Phase ….
Step 6: Approval of Project Management Plan

Stakeholders Review
Present the project management plan to key stakeholders for review.

Feedback and Revisions
Incorporate feedback and make necessary revisions to the plan.

Final Approval
Obtain formal approval to proceed with the execution phase.

PLC – Planning Phase

Concluding Remarks of Planning Phase

The planning phase culminates in a detailed project management plan that acts as a blueprint for project execution.

This plan ensures all team members and stakeholders are on the same page and fully understand their roles, the project timelines, the scope of work, and the measures in place to manage risks and communications.

With this solid plan, the project is well-prepared to move into the execution phase, where the actual development and building of the mobile banking application will begin.

Execution Phase

PLC – Execution Phase

Execution Phase

- The execution phase is where most of the work on the project happens—it's about delivering on the commitments made in the project plan.
- Ensuring that human, financial, and physical resources are secured and managed effectively.
- The tasks identified during the planning phase are carried out, monitored, and controlled based on performance against the project management plan.
- Ensuring that project deliverables meet the quality standards set out in the planning phase.
- Keeping stakeholders engaged and informed throughout the project lifecycle.

PLC – Execution Phase

An Example of an Execution Phase ….

Execution Phase: Mobile Banking Application Development Project

Step 1: Kick-off Execution

Kick-off Meeting

Conduct a formal kick-off meeting with the project team, stakeholders, and any third-party vendors to mark the start of project execution. Reiterate the project goals, timeline, key milestones, and individual roles and responsibilities.

Step 2: Develop and Integrate

App Development

The development team starts coding according to the design specifications.

They work in iterations, allowing for incremental building and testing of the app's features.

Integration

Simultaneously, the IT team works on integrating the app with existing bank systems, such as databases for customer data, transaction systems, and authentication servers.

PLC – Execution Phase

An Example of an Execution Phase ….

Step 3: Quality Assurance and Testing

Unit Testing

Developers perform unit testing to ensure that individual components of the app work correctly.

Integration Testing

Test integration points between the app and existing bank systems

User Acceptance Testing (UAT)

Engage a group of beta testers or actual end-users to test the app and provide feedback. This helps ensure that the app meets user requirements and is user-friendly.

Security and Compliance Testing

Conduct thorough security checks and compliance verifications to ensure the app meets all relevant standards and regulations.

PLC – Execution Phase

An Example of an Execution Phase ….

Step 4: Performance Management

Monitoring and Controlling

Use project management tools to track progress against the project plan. Regular status meetings help monitor milestones, manage resources, adjust timelines, and resolve issues.

Quality Control

Implement quality control processes to adhere to the quality management plan. Regular reviews and audits ensure deliverables meet the quality standards established in the planning phase

PLC – Execution Phase

An Example of an Execution Phase ….

Step 5: Stakeholder Communication and Reporting

Regular Updates

Provide stakeholders with regular updates on progress, challenges, and any changes. Use the communication plan as a guide for frequency and methods of communication.

Issues Management

Record and manage any issues that arise, ensuring they are addressed promptly and do not affect the project timeline or deliverables.

PLC – Execution Phase

An Example of an Execution Phase ….

Step 6: Documentation

Technical Documentation

Maintain comprehensive documentation of the app's architecture, code, and integration interfaces.

User Documentation

Develop user manuals, FAQs, and online help resources to assist users in using the app effectively.

PLC – Execution Phase

An Example of an Execution Phase ….

Step 7: Training

Staff Training

Conduct training sessions for bank staff and customer service representatives.

Training includes how to use the app, how to troubleshoot common issues, and how to assist customers effectively.

Stakeholders Demonstrations

Organize demonstrations for stakeholders to showcase app functionality and features, ensuring it aligns with their expectations and gathers further feedback.

PLC – Execution Phase

Concluding Remarks of Execution Phase

The execution phase is where the planned activities are turned into tangible deliverables.

It requires effective coordination of all project components—people, processes, and technology—to ensure the mobile banking application is developed successfully, tested thoroughly, and ready for deployment.

As this phase wraps up, the project moves into the monitoring and controlling processes to ensure everything aligns with the project management plan before heading to the final closure phase.

Closure Phase

PLC – Closure Phase

Closure Phase

- This final phase ensures that the project is brought to an orderly end.
- Ensuring that all aspects of the project are completed, confirmed by the customer, and formally closed.
- Delivering the final product to the customer, and handing over project documentation.
- Evaluating what went well and what didn't, documenting lessons learned to improve future projects.
- Disbanding the project team and releasing resources back to the organization or on to other projects.

PLC – Closure Phase

An Example of Closure Phase ….

Closure Phase: Mobile Banking Application Development Project

Step 1: Completion and Handover

Final Testing and Approval

Conduct the final round of testing, including load testing and security testing, to ensure the app is robust and secure.

Obtain final approval from the stakeholder committee, which includes senior bank management and IT leads.

App Deployment

Deploy the mobile banking application to the production environment.

This includes making the app available for download in app stores and ensuring it integrates seamlessly with live databases and transaction systems.

Customer Rollout

Initiate a phased rollout to customers, starting with a small group to gather initial feedback and ensure smooth operation before a full rollout.

PLC – Closure Phase

An Example of Closure Phase ….

Step 2: Documentation and Deliverables

Project Documentation

Compile all project documentation, including design documents, test records, compliance approvals, and deployment guides, to hand over to the bank's IT operations team.

User and Maintenance Guides

Provide comprehensive user guides for customers and maintenance and support manuals for the IT operations team.

PLC – Closure Phase

An Example of Closure Phase ….

Step 3: Project Review

Post-Implementation Review

Conduct a meeting with the project team and key stakeholders to review the project's successes and challenges. Discuss what worked well and what could be improved for future projects.

Lessons Learned

Document lessons learned during the project, focusing on insights that can help improve future projects. This includes aspects related to project management processes, technology implementation, stakeholder engagement, and risk management.

PLC – Closure Phase

An Example of Closure Phase ….

Step 4: Release Resources

Resources Release

Release project resources, including staff, contracted vendors, and technologies.

Ensure that all contractual obligations are fulfilled, final payments are processed, and official contracts are closed.

Team Recognition

Recognize the efforts of the project team through formal acknowledgments, rewards, or celebration events.

This helps in team motivation and retention.

PLC – Closure Phase

An Example of Closure Phase ….

Step 5: Final Reporting

Final Project Report

Prepare a final project report summarizing the project outcomes, budget usage, timeline adherence, and scope achievement.

Include performance metrics and outcomes of the business objectives, like user adoption rates and operational efficiencies gained.

Stakeholder Presentation

Present the final report to senior management and other key stakeholders.

Provide insights into the project's impact on the business and potential next steps for further enhancements or additional features.

PLC – Closure Phase

An Example of Closure Phase ….

Step 6: Archiving

Archiving Files

Archive all project-related documents and materials in a secure environment.

This ensures that all project information is available for future reference, audits, or any subsequent projects.

PLC – Closure Phase

An Example of Closure Phase ….

Remarks

The closure phase formally concludes the project by ensuring all project deliverables have been satisfactorily completed and accepted.

It also involves evaluating the project process, capturing lessons learned, and effectively releasing project resources.

This phase not only wraps up the project in a structured manner but also sets a foundation for ongoing support and future initiatives.

With the successful closure of the project, the mobile banking application is now fully operational, with ongoing operations handed over to the bank's IT and operational teams.

PROJECT SCHEDULING TECHNIQUE

Introduction

Important Questions During Project Implementation

- What activities can produce expected project outputs?
- What is the sequence of these activities?
- What is the time frame for these activities?
- Who will be responsible for carrying out each activity?

Introduction

The following methods may be used to answer the above questions:

- Gantt chart
- Project Evaluation and Review Techniques (PERT)
- Critical Path Method (CPM) or Network analysis.
- Simple formats e.g., Bar charts, timelines, milestones etc.

What is Project Scheduling?

Project Scheduling:

- is the tool that communicates;
- what work needs to be performed
- which resources of the organization will perform the work
- the timeframes in which that work needs to be performed.

- It reflects all of the work associated with delivering the project on time.

- Without a full and complete schedule, the project manager completes the project effort in terms of cost and resources necessary to deliver the project.

Project Scheduling …

- A project is a collection of tasks that must be completed in minimum time or at minimal cost.
  - what work needs to be performed
  - which resources of the organization will perform the work
  - the timeframes in which that work needs to be performed.

- **Objectives of Project Scheduling**
  - Completing the project as early as possible by determining the earliest start and finish of each activity.
  - Calculating the likelihood that a project will be completed within a certain time period.
  - Finding the minimum cost schedule needed to complete the project by a certain date.

Project Scheduling …

- Scheduling *means the estimation of time and resources required to complete activities and organize them*.

- The following tools can be used in project scheduling
  - Critical Path Method
  - PERT chart
  - Gantt chart

Network Planning &
Network Diagrams

Network planning

- Network planning is the process of organizing a network to ensure efficient communication and data transfer.

- Network analysis incorporates a variety of techniques used to help plan and manage the schedule of interrelated activities.

- Network-Based Techniques are project management methods that involve representing project activities and their interdependencies using graphical diagrams.

- Useful in project management for:
  - Scheduling
  - Monitoring project
  - Controling
  - Resource allocation
  - Reviewing costs

- Common for complex projects involving hundreds of activities

Network planning …

- A **network diagram**: This is a diagrammatic representation of the individual project activities in their logical & practical sequence that establishes a relationship.

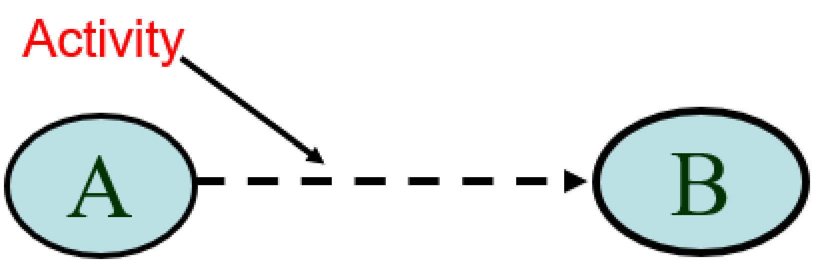
- **An activity:** This is the task or part of a work that requires *time* and *resources* for its implementation (the opposite is a dummy activity).

- **An event:** Starting or ending point of an activity (tail and head event

Network planning …

Important Network Notations

- Network is a diagrammatic illustration of the relationship btw a group of activities. A network is made of two things:
  - Activity presented as an arrow
  - Events presented as a node



Network planning …

Rules for Drawing Network Diagram

1. Each project activity should be presented by only one arrow.

2. An activity can begin only when all its successors are done

3. No two activities can be identified by same tail and head

4. All activity flows must be shown as moving from left to right; so is the numbering of events.

5. Introduce as few dummy activities as may be necessary

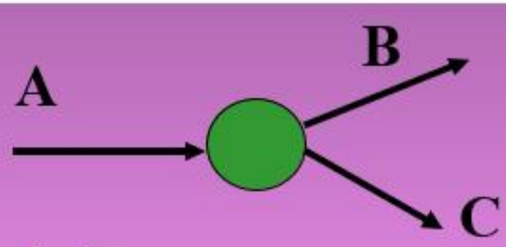
Network planning ...

Rules for Drawing Network Diagram

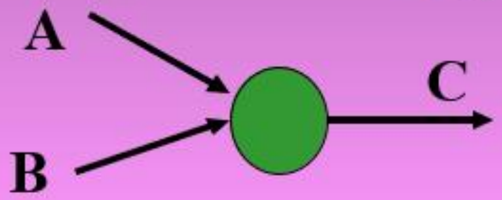
6. There should be no circuits or loops in network (nor circulating)
7. A network should have only one initial and one terminal
8. Proper activity precedence must be followed – presented in practical and logical sequence.
9. Network arrows are not drawn to scale but their direction is important. (length of an arrow ≠ the duration of the activity)

Network planning …

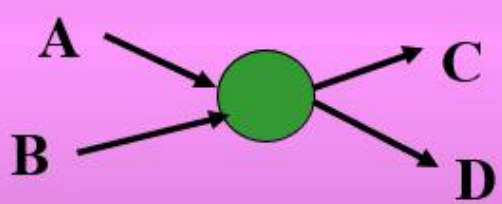
Situations in Network Diagram



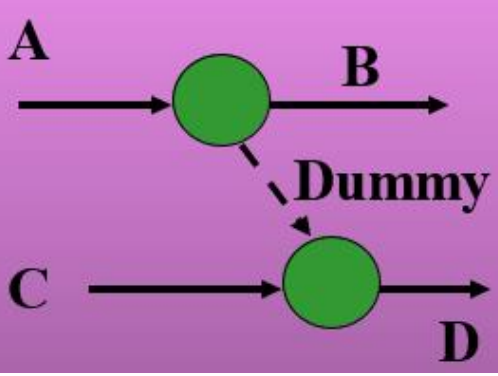
A must finish before either B or C can start



both A and B must finish before C can start



both A and B must finish before either of C or D can start



A must finish before B can start

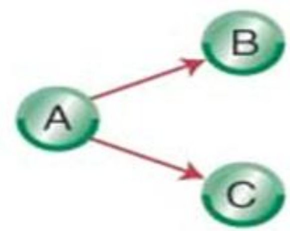
both A and C must finish before D can start

Network Diagrams

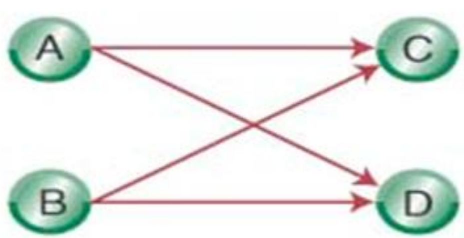
Activity-on-Node (AON)

- Uses nodes to represent the activity
- Uses arrows to represent precedence relationships

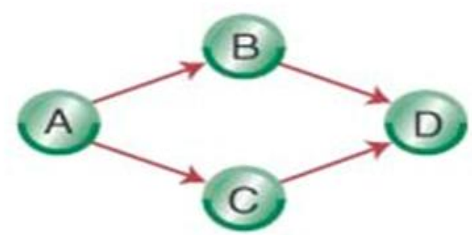
a. Activity A precedes activity B, which precedes activity C



b. Activity A must be completed before activities B and C can begin.



c. Activities A and B must both be completed before activity C or D can begin.



d. Activities B and C can begin once activity A has been completed; activity D cannot begin until both B and C are completed.

What is the Critical Path Method (CPM)?

- Is the network-based technique used which shows the sequence of scheduled activities that determines the duration of the project.

Or

- Is the longest sequence of tasks in a project plan that must be completed on time in order for the project to meet its deadline

- A delay in any task on the critical path, delays the whole project

- Although many projects have only one critical path, some projects may have multiple critical paths.

Network-Based Techniques

CPM …

- The critical path is important when managing a project because it identifies all the tasks needed to complete the project
- Then determine the tasks that must be done on time, those that can be delayed if needed (due to the triple constraint of time, cost, and scope), and how much float you have.
- With this knowledge, you can prioritize, set realistic deadlines, and assign critical tasks to your most capable team members.

STEPS in CPM

CPM …

1. Define/list all project activities

2. Estimate the activity durations.

3. Identify dependencies (interrelationships) between activities.

4. Create a network diagram.

5. Analyze the network by calculating the critical path and determining project duration. Calculate the Early Start (ES) Time , and Early Finish (EF) Time
   - ES or EST is the term used in the CPM to describe the earliest time that an activity can begin

STEPS in CPM …

- In project management, an activity is a set of actions that are performed to produce a project's deliverables.

- Activities are the smallest parts of a project that are used for planning, tracking, and control.

- The total project is subdivided into activities and each activity is given an alphabetical symbol/code.

- This involves a detailed delineation of the activities to be performed to complete the project.

STEPS in CPM …

Step1: Prepare the list of activities

- Take an example of activities in Table 01.

Table 01: List of activities

S/N	Activity	Symbol / Code
1	Site Selection	A
2	Digging well	B
3	Laying field channels	C
4	Procurement of pump	D
5	Installation of pump	E
6	Test run	F

STEPS in CPM …

Step 2: Define the interrelationship among the activities (activity Sequence).

- The relationship among the activities could be defined by specifying the preceding and succeeding activity.

- The preceding activity for an activity is its immediate predecessor, i.e. the activity that needs to be completed before the start of the new activity.

Table 2: Activity Sequence

S/N	Activity	Symbol/code	Preceding activity
1	Site selection	A	-------
2	Digging well	B	A
3	Laying field channels	C	B
4	Procurement of pump	D	A
5	Installation of pump	E	B, D
6	Test runs	F	C, E

STEPS in CPM …

Step 2: Define the interrelationship among the activities ….

- In the given example, “the selection of the site” precedes the digging of the well.

- In other words, “the site” needs to be selected before digging the well. Thus the activity “Selection of the site” becomes the proceeding activity to the activity of “Digging the well”

- Succeeding activity is the one that immediately starts after completion of the activity. “**Digging well**” is the succeeding activity to “**Selection of the site**”.

STEPS in CPM …

Step 3: Estimate Activity Duration

- The activity time is the time that is actually expected to be expended in carrying out the activity.

Table 03: Activity Time Estimates

S/N	Activity	Symbol/Code	Preceding Activity	Estimated Time
1	Site selection	A	-----	7
2	Digging well	B	A	3
3	Laying field channels	C	B	15
4	Procurement of pump	D	A	7
5	Installation of pump	E	B, D	3
6	Test run	F	C, E	2

STEPS in CPM …

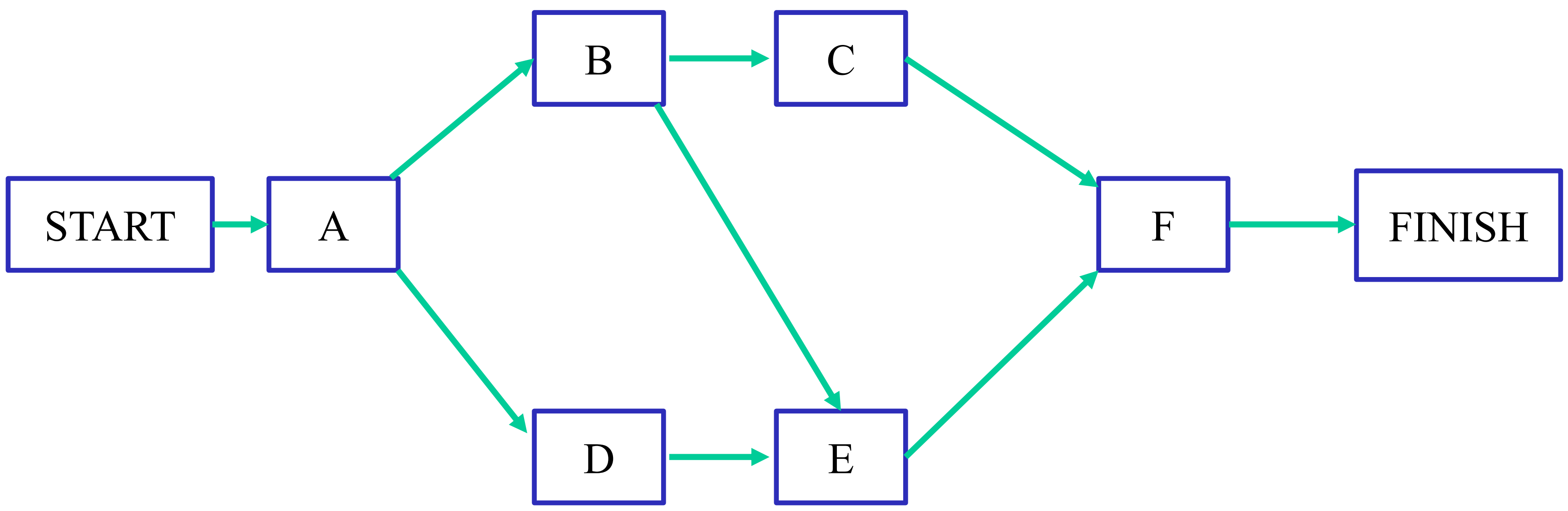
Step 4: Draw the Network Diagram

- The activities identified and their dependencies aid you in drawing the critical path analysis chart (CPA)

- Is also known as the network diagram.

- The network diagram is a visual representation of the order of your activities based on dependencies.

Step 4: Draw the Network Diagram …



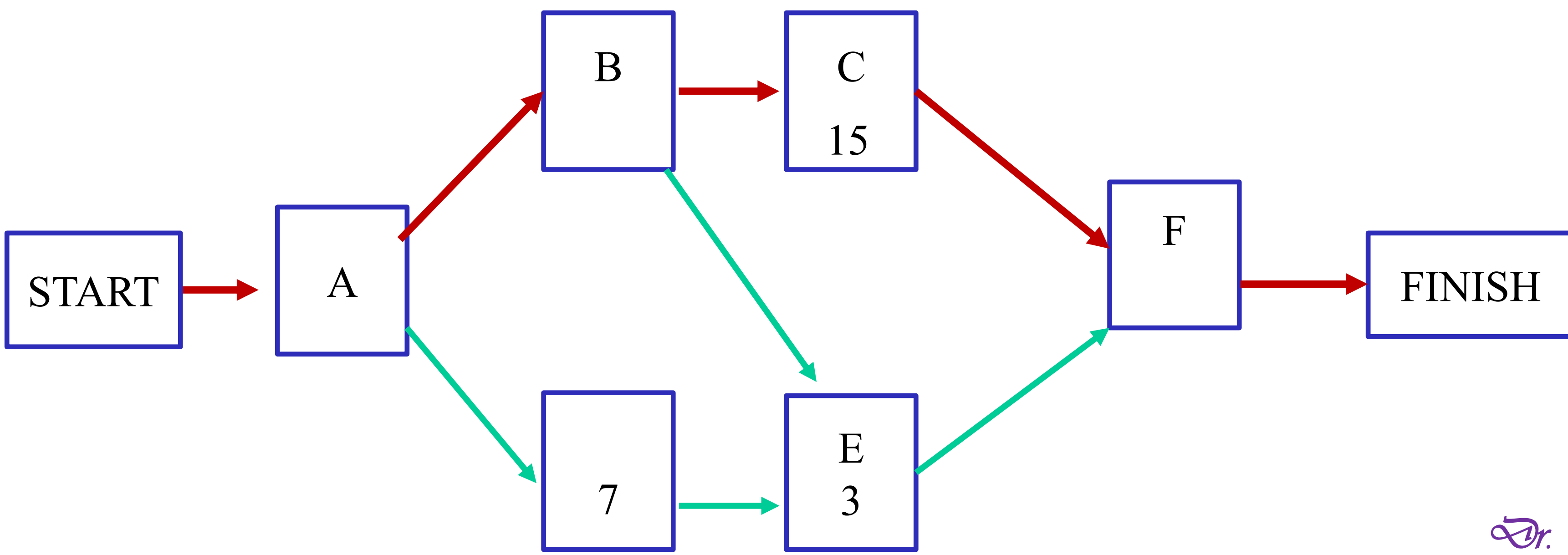
STEPS in CPM …

Step 5: Identify the Critical Path

Two ways you can now identify the critical path

1. Eyeball the network diagram and simply identify the longest path throughout the network: - the longest sequence of activities on the path.

- Look for the longest path in terms of the longest duration in days, not the path with the most boxes or nodes



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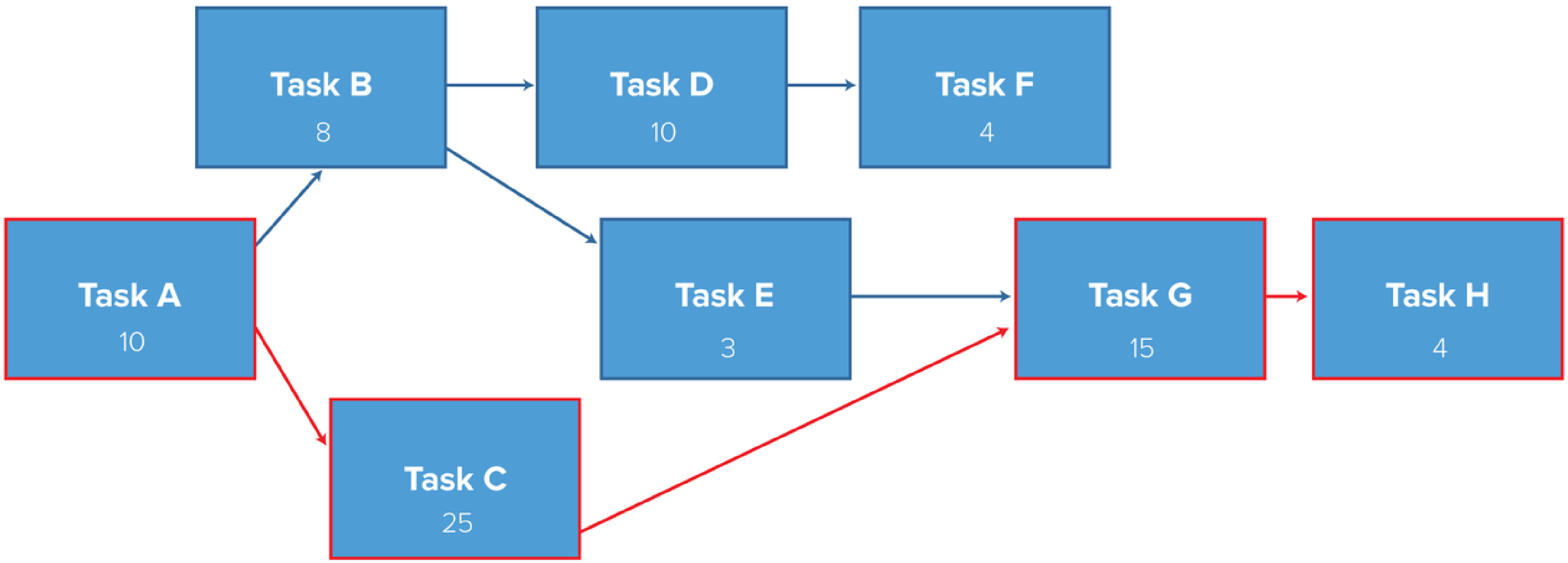
STEPS in CPM …

Step 5: Identify the Critical Path



STEPS in CPM …

Step 5: Identify the Critical Path

Two ways you can now identify the critical path

2. Secondly Identify critical activities with the Forward Pass/Backward Pass technique.

- Enable you to identify the earliest start (ES) and Earliest finish(EF) times, and the latest start (LS) and (LF) finish times for each activity

Importantly Things in CPM

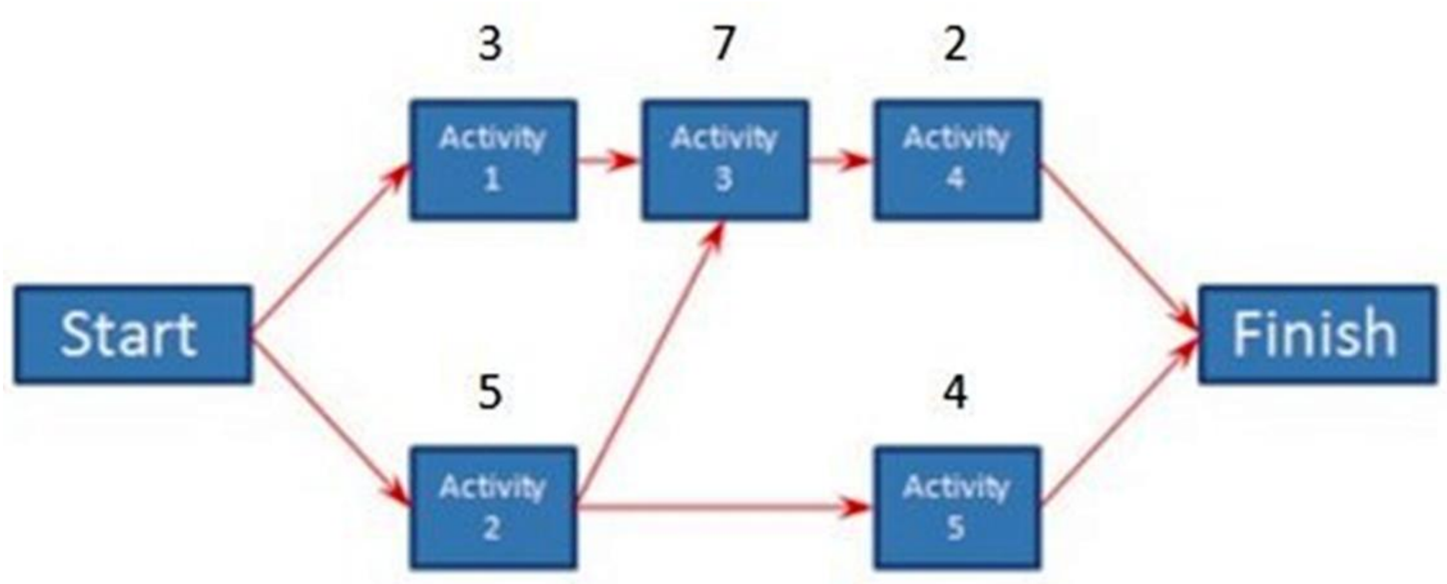
- The CPM has four key elements
  - Critical Path Analysis
  - Float determination
  - Early Start & Early Finish Calculation
  - Late Start & Late Fish Calculation

- CP is the sequence of activities with the longest duration

- Delay in one activity leads to a delay in the whole project

Importantly Things in CPM

- Lets look at some critical path examples to bring out the concepts
- Suppose a project has the following activities.



- Each activities duration is above each node

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Importantly Things in CPM

- To get the critical path we add the duration of each node to determine its total duration
- CP is the one with the longest duration
- In the project above there are 3 paths thru the project

Critical Path

Start → Activity 1 → Activity 3 → Activity 4 → Finish   3 + 7 + 2 = 12

Start → Activity 2 → Activity 3 → Activity 4 → Finish   5 + 7 + 2 = 14

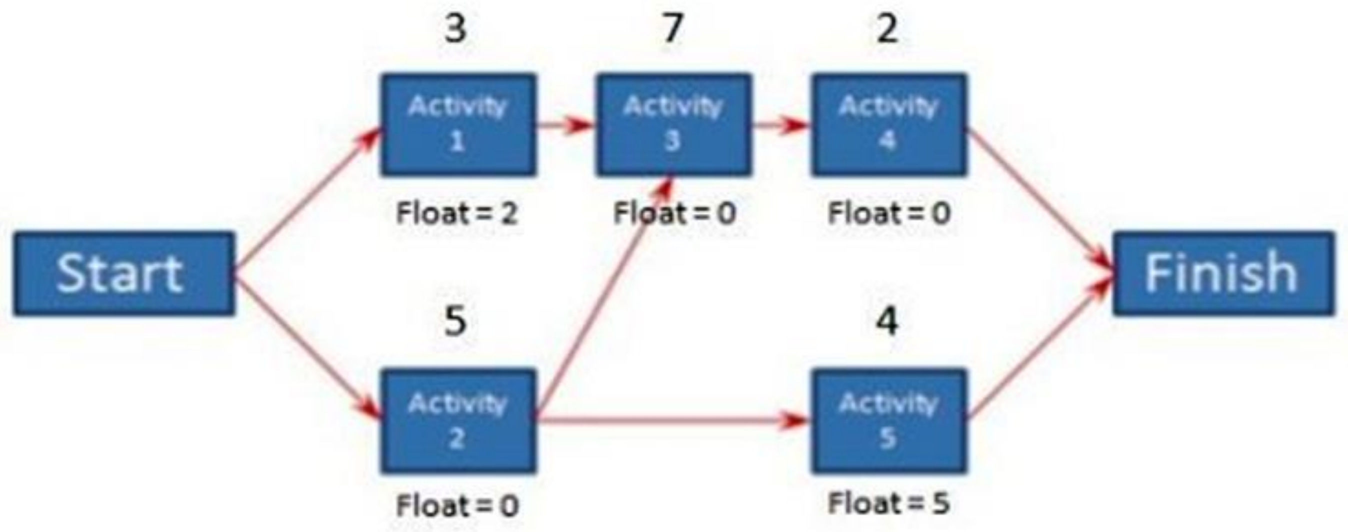
Start → Activity 2 → Activity 5 → Finish   5 + 4 = 9

- The Critical one is path 2

Determining the Float

Determining the Float

- Float is the amount of time (days) an activity can slip before it causes the project to be delayed
- This is sometimes called Slack
- Start with activities on the Critical Path.
- Each of those activities has a Float of 0, meaning if any of the activities slip, the project is delayed



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Determining the Float …

- Take the next longest path - Subtract its duration from the duration of the critical path to get float for each of those activities

- Continue doing the same for each subsequent longest path till each activity float is determined

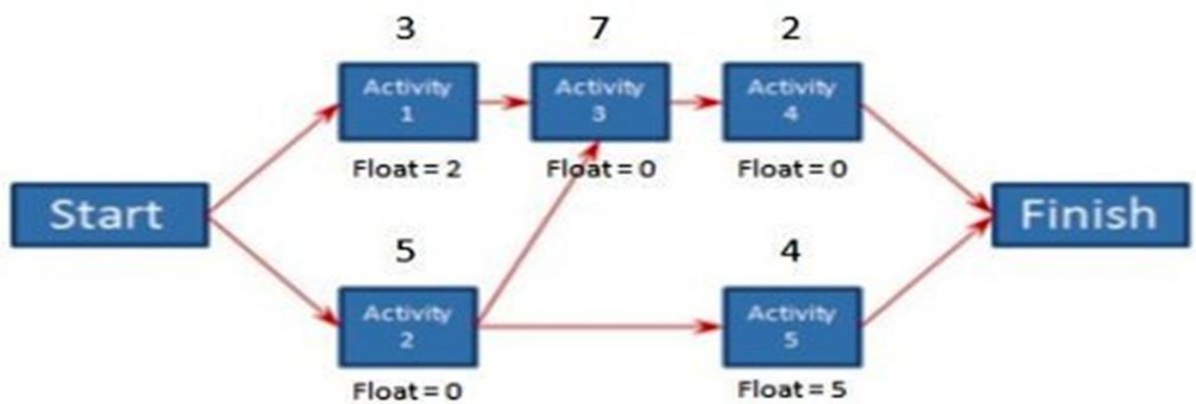
- If an activity is on two paths, its float will be based on the long path that it belongs to.

- Float Can also be computed as:

  Total float = TF = Latest start – Earliest Start

Determining the Float …

- In the case of our CP diagram Activities 2,3 and 4 are on a critical path so have float 0.
- The next longest path is activities 1,3 & 4. Since Activities 3 & 4 are on CP their float remains 0.
- For activity 1 float will be Duration of Critical path - duration of this path = **14 -12 =2**.
- The next path is activities 2 & 5. Activity 2 is on CP so float =0
- Activity 5 has a float of **14-9 =5.**
- Means so long Activity 5 doesn't slip more than 5 days it won't cause a delay in the project.



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EARLY START & EARLY FINISH

Early Start & Early Finish

- CPM includes a technique called Forward pass
- Used to determine *the earliest date an activity can start and the earliest date it can finish*.
- These dates are valid as long as all prior activities in that path started on their earliest start date and didn't slip.
- Starting with CP the Early Start (ES) of Activity 1 = 1
- The early finish (EF) of an activity is its ES plus its duration minus one.

Early Start & Early Finish

- Using our earlier example, Activity 2 is the first activity on the critical path:
  - ES = 1,
  - EF = 1 + 5 -1 = 5.

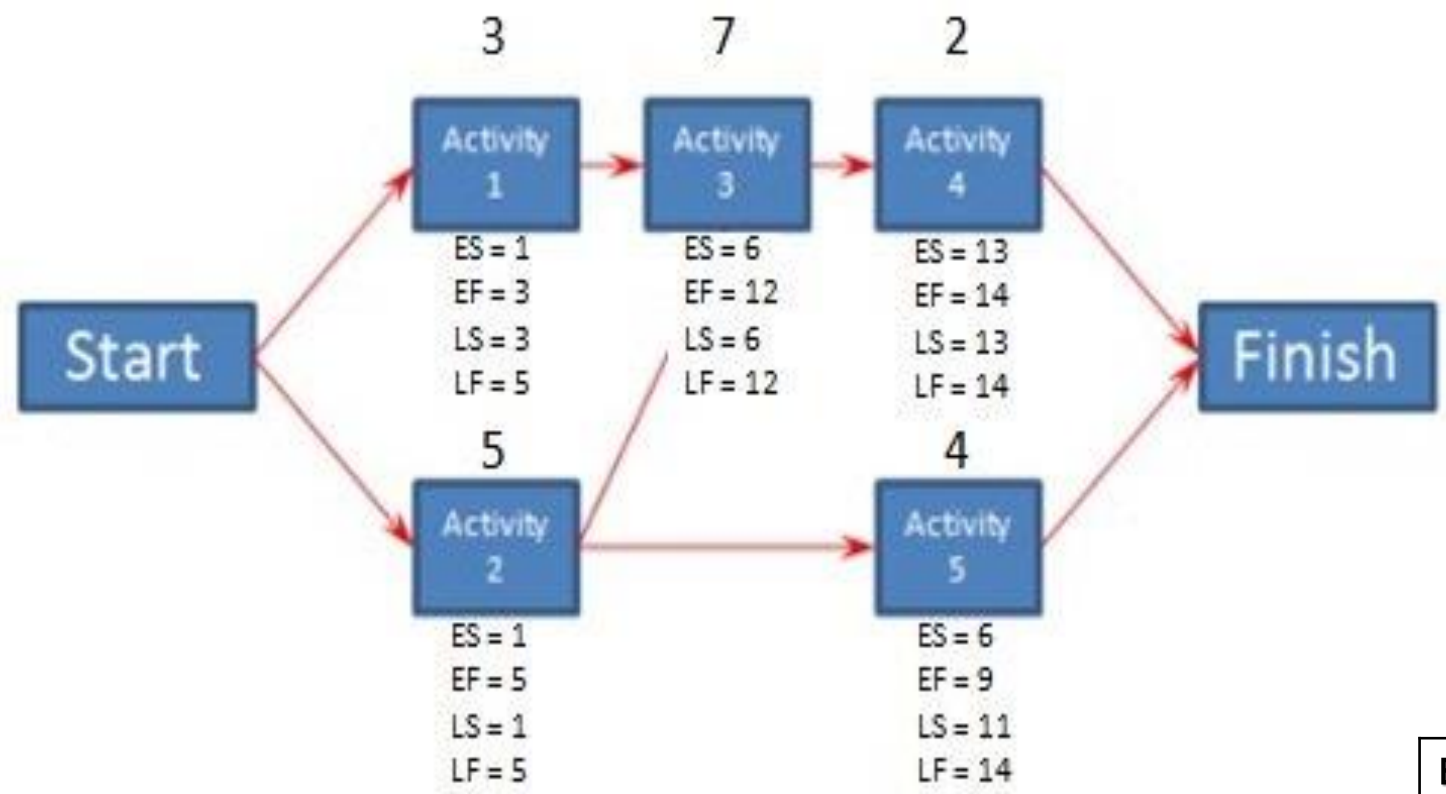
- Activity 3 its ES is Previous activities EF+1

- Activity 3 ES = 5 +1 = 6

- Its EF is calculated as before;
  - EF = 6+7-1 =12

- To calculate the ES of an Activity with more than one predecessor, you will use the activity with the latest EF.

Early Start & Early Finish



ES	ACT	EF
LS	DUR	LF

LATE START & LATE FINISH

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Late Start & Late Finish

- The Backward Pass is a CPM technique you can use to determine the latest date an activity can start and the latest date it can finish before it delays the project .

  - Start with CP but begin from last activity of the path
  - The Late Finish (LF) for the last activity in every path is the same as the last activity's EF in the critical path.
  - The Late Start (LS) is the LF - duration + 1.
  - For E.g. Activity 4 is the last activity on the critical path. Its LF is the same as its EF, which is 14.
  - To calculate the LS, subtract its duration from its LF and add one. LS = 14 - 2 + 1 = 13.

Late Start & Late Finish

- The next Activity in the critical path is Activity 3.
- Its LF is equal to Activity 4 LS - 1.
- Activity 3 LF = 13 -1 = 12.

- It's LS is calculated the same as before by subtracting its duration from the LF and adding one.
- Activity 3 LS = 12 - 7 + 1 = 6.

- CPM is an important tool for managing project schedules
- With large projects with many activities PM software easily does this quickly

Summary

- **Free slack** or **free float** is the amount of time an activity can be delayed without delaying the early start of any immediately following activities

- **Total slack** or **total float** is the amount of time an activity may be delayed from its early start without delaying the planned project finish date

- A **forward pass** through the network diagram determines the early start and finish dates

- A **backward pass** determines the late start and finish dates

Differences between PERT & CPM

BASIS	PERT	CPM
Meaning	PERT is a Project management technique, used to manage uncertain activities of a project	CPM is a statistical technique of project management that manages well defined Activities of a project.
What is it?	A technique of planning and control of time	A method to control cost and time.
Orientation	Event-oriented	Activity-oriented
Usage	Used in Research & Development project – activities are unpredictable	Used in Construction project – Activities are Predictable
Model	Probabilistic Model – cant estimate completion time	Deterministic Model
Focuses on	Time - meeting time target	Time-cost trade-off – optimize cost min time duration
Estimates	Three time estimates -Optimistic, most likely & pessimistic time	One time estimate
Suitable for	projects needing High precision time estimate	Projects needing Reasonable time estimate
Management of	Unpredictable Activities	Predictable activities
Nature of jobs	Non-repetitive nature	Repetitive nature